

BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



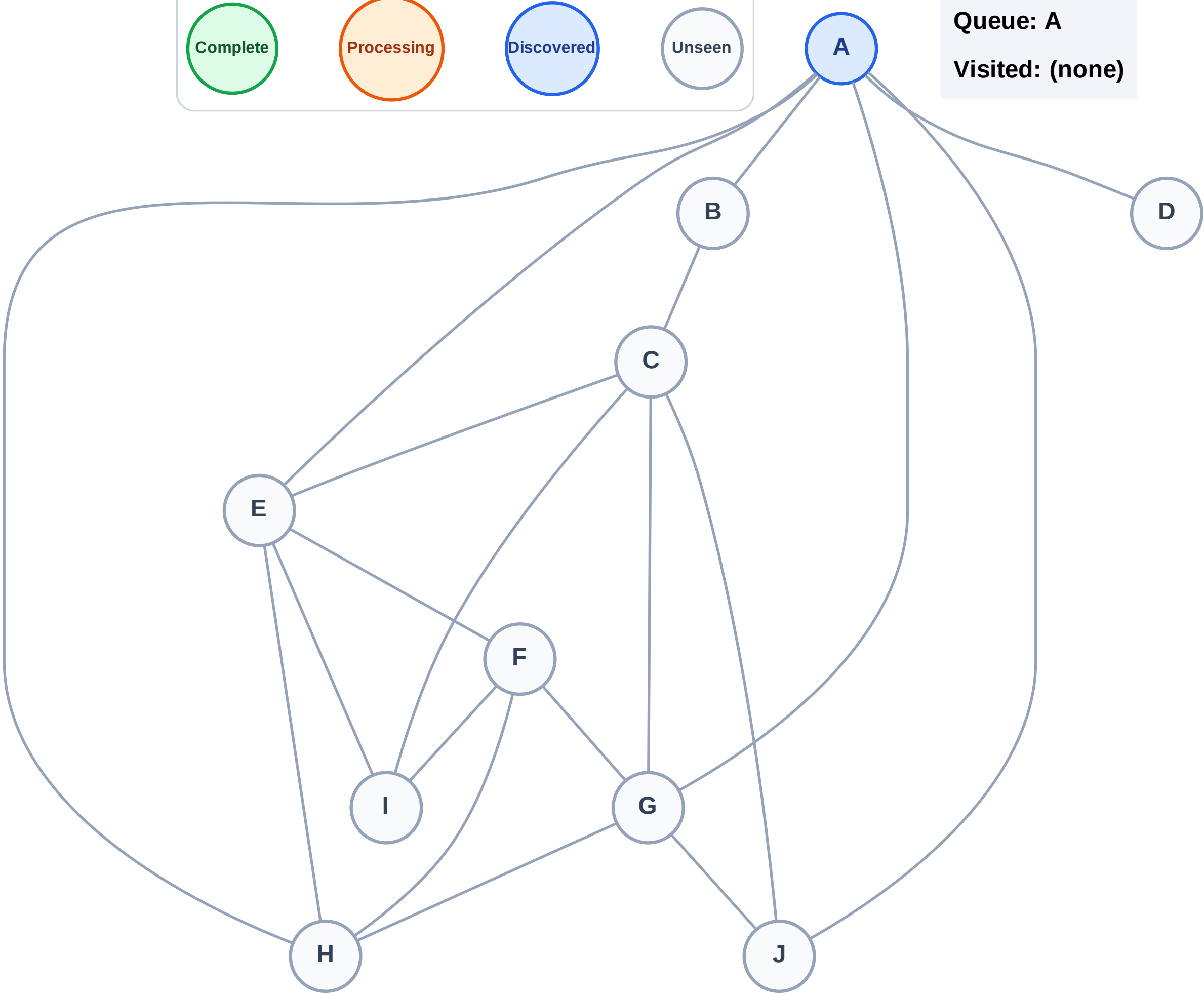
Discovered



Unseen

Queue: A

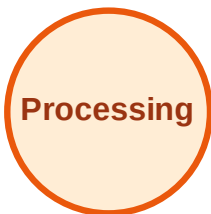
Visited: (none)



BFS Traversal

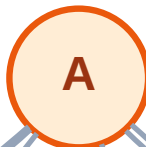
Step 2: Process node A

Legend



Queue: (empty)

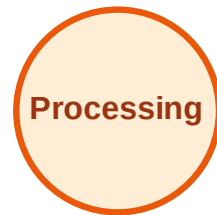
Visited: (none)



BFS Traversal

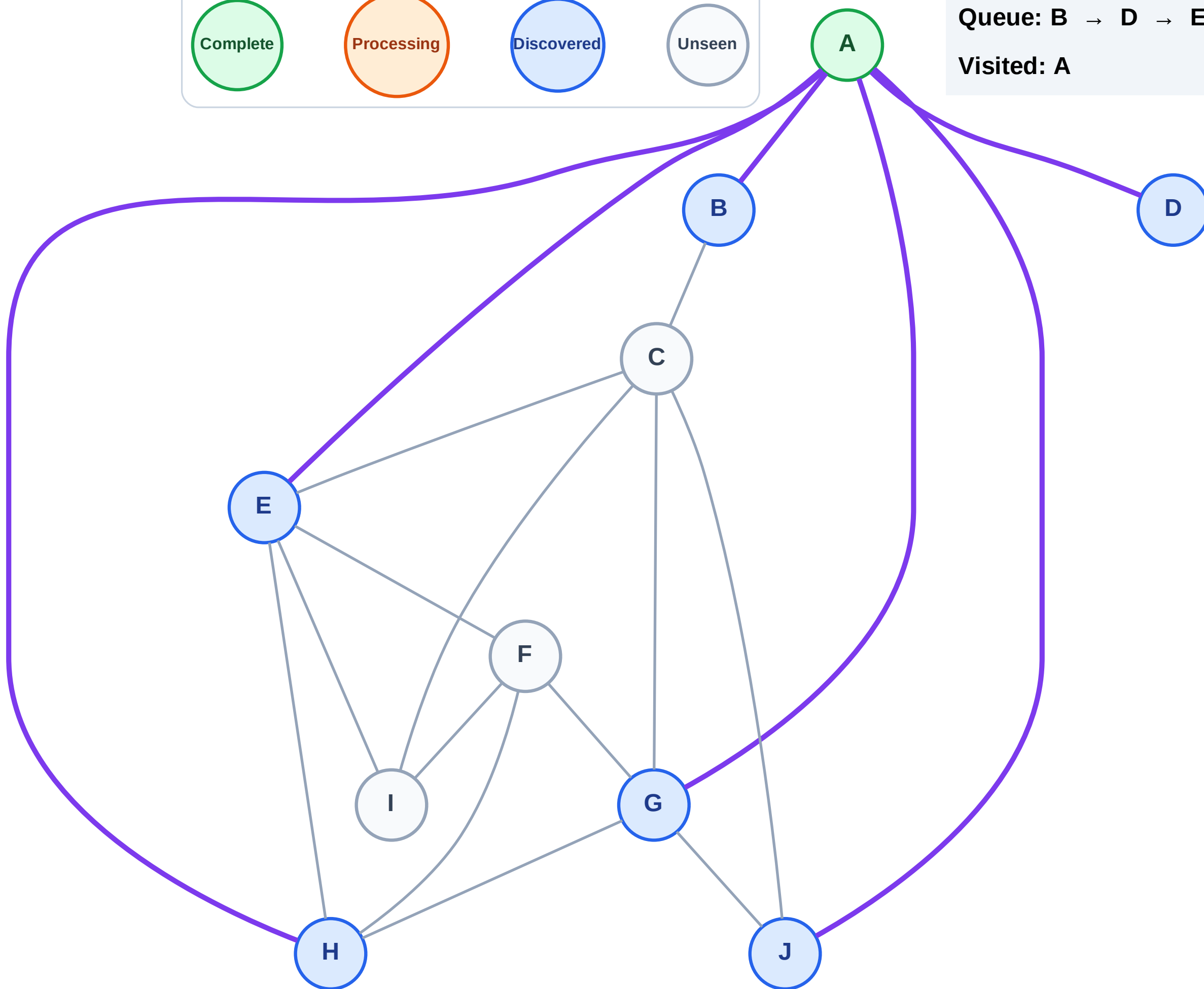
Step 3: Discover B, D, E, G, H, J from A

Legend



Queue: B → D → E → G → H → J

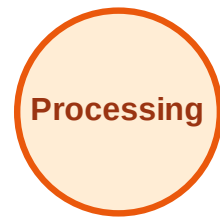
Visited: A



BFS Traversal

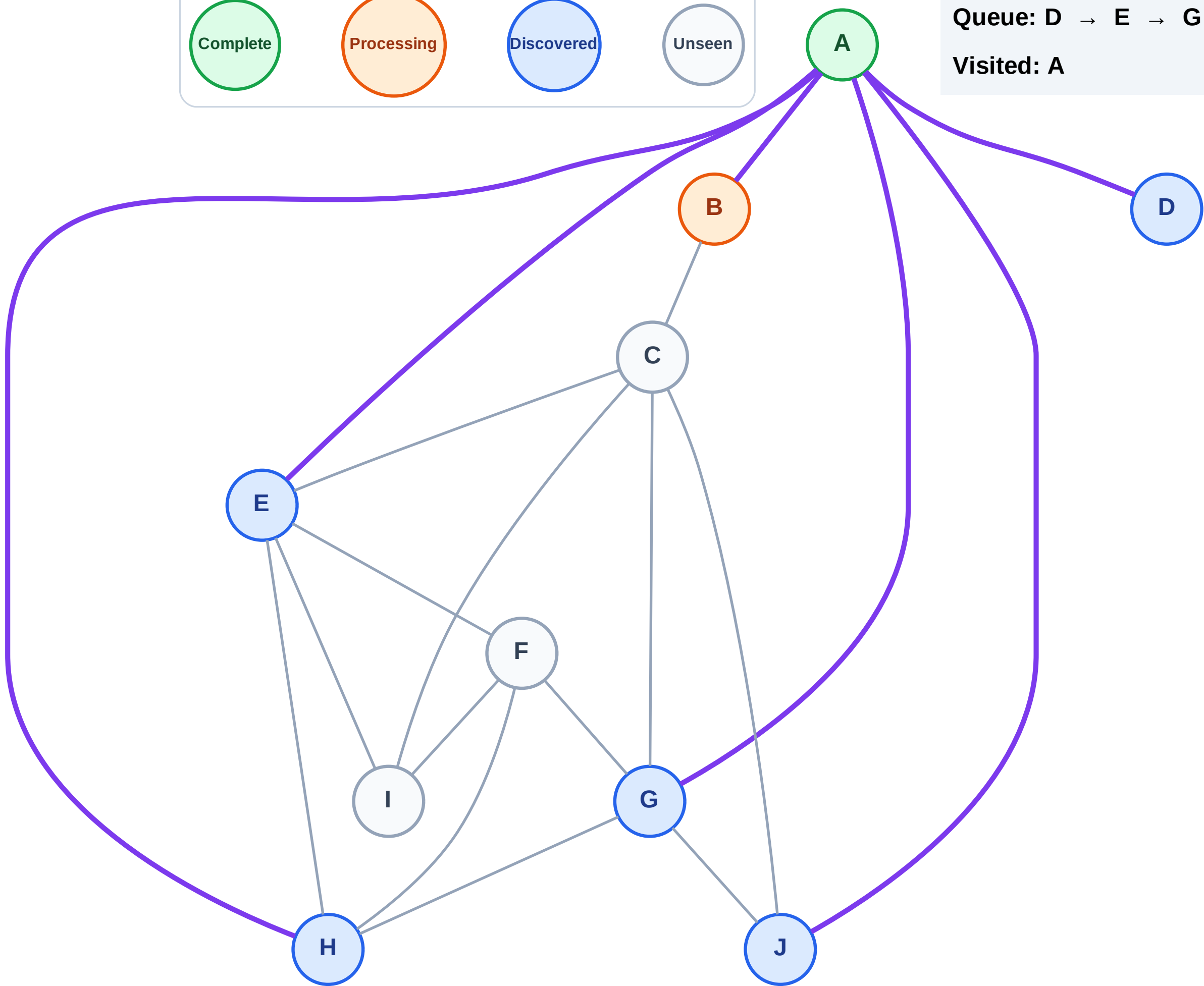
Step 4: Process node B

Legend



Queue: D → E → G → H → J

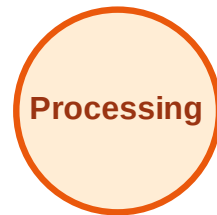
Visited: A



BFS Traversal

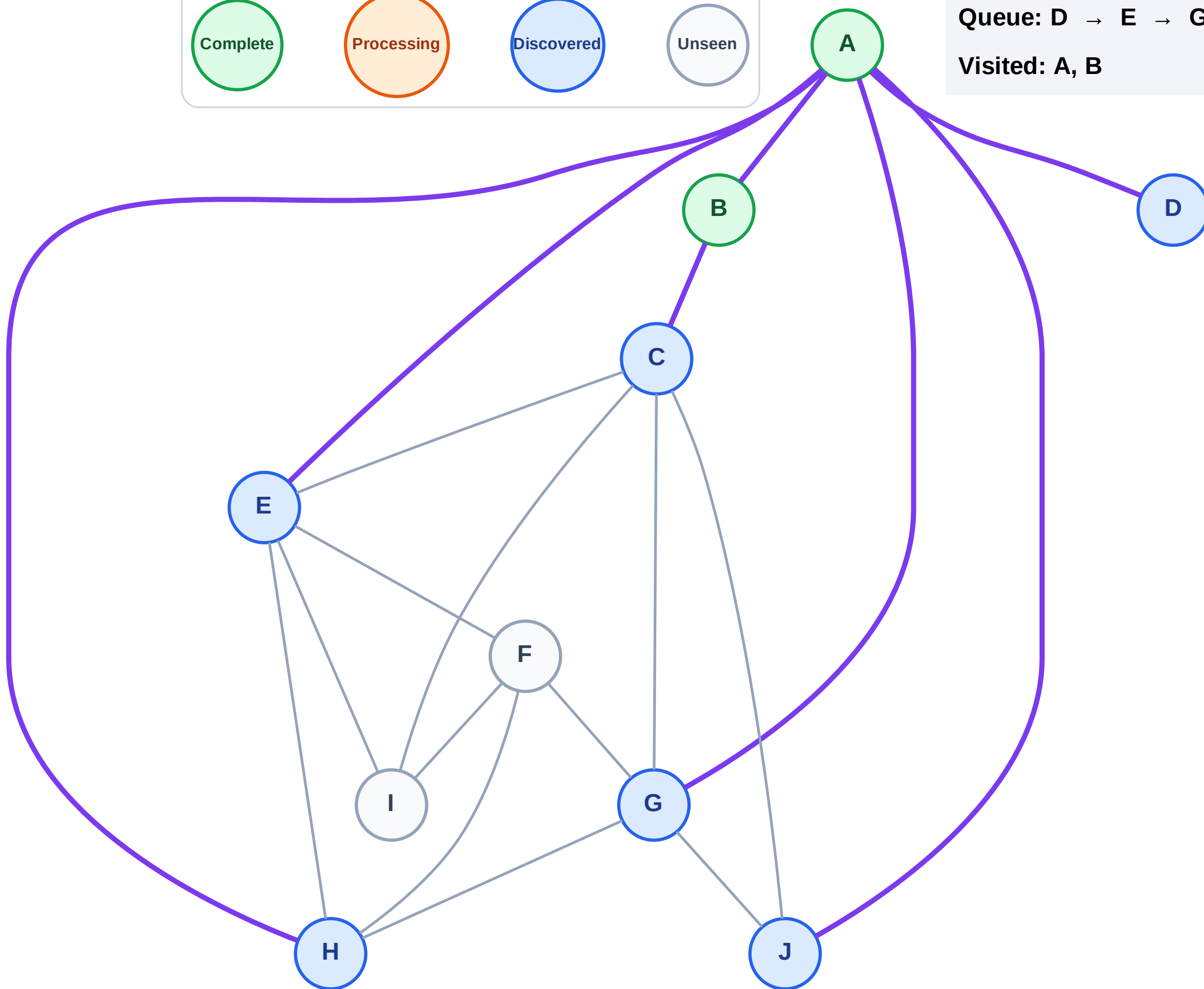
Step 5: Discover C from B

Legend



Queue: D → E → G → H → J → C

Visited: A, B



Step 6: Process node D

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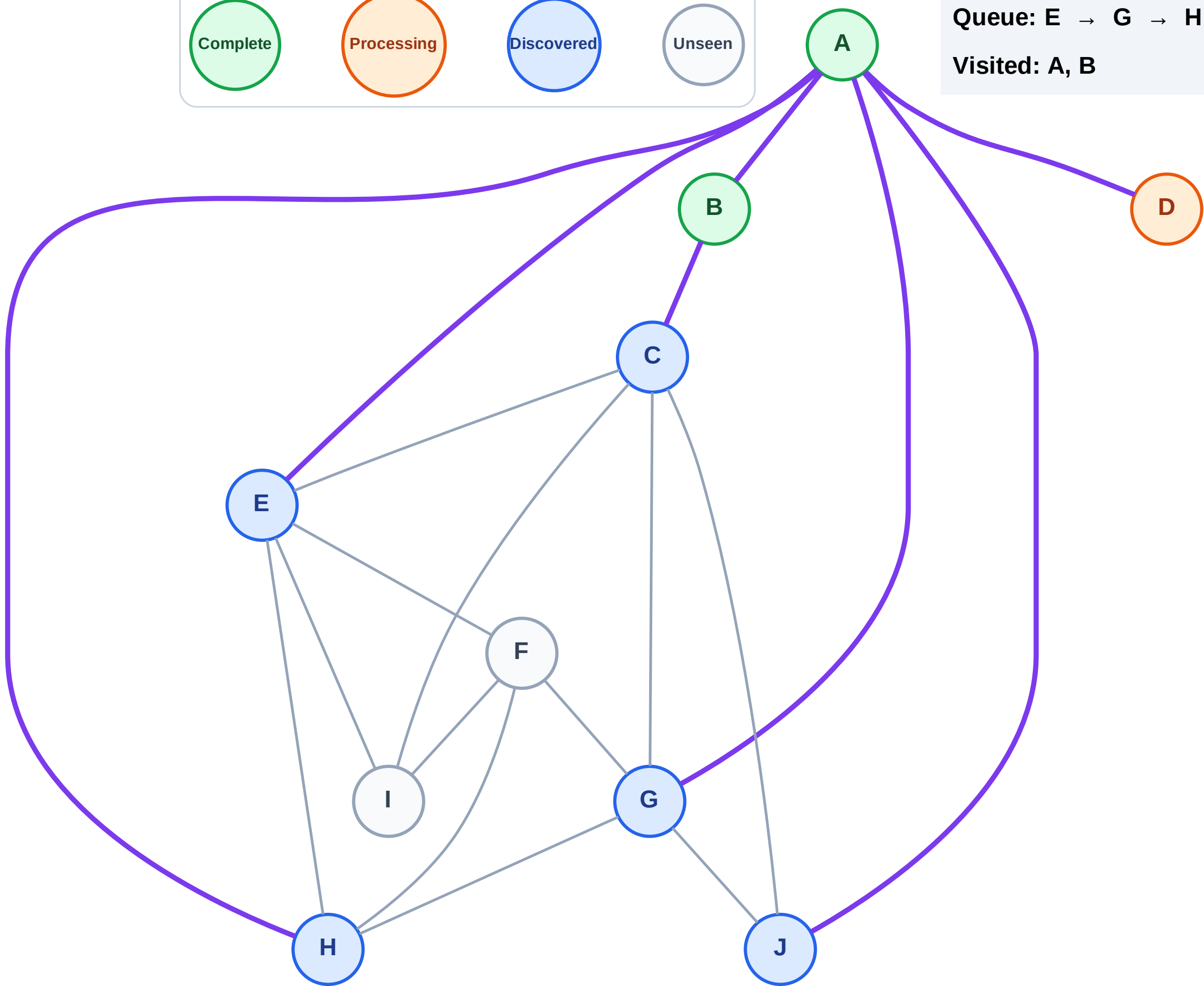
Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Unseen

Visited: A, B



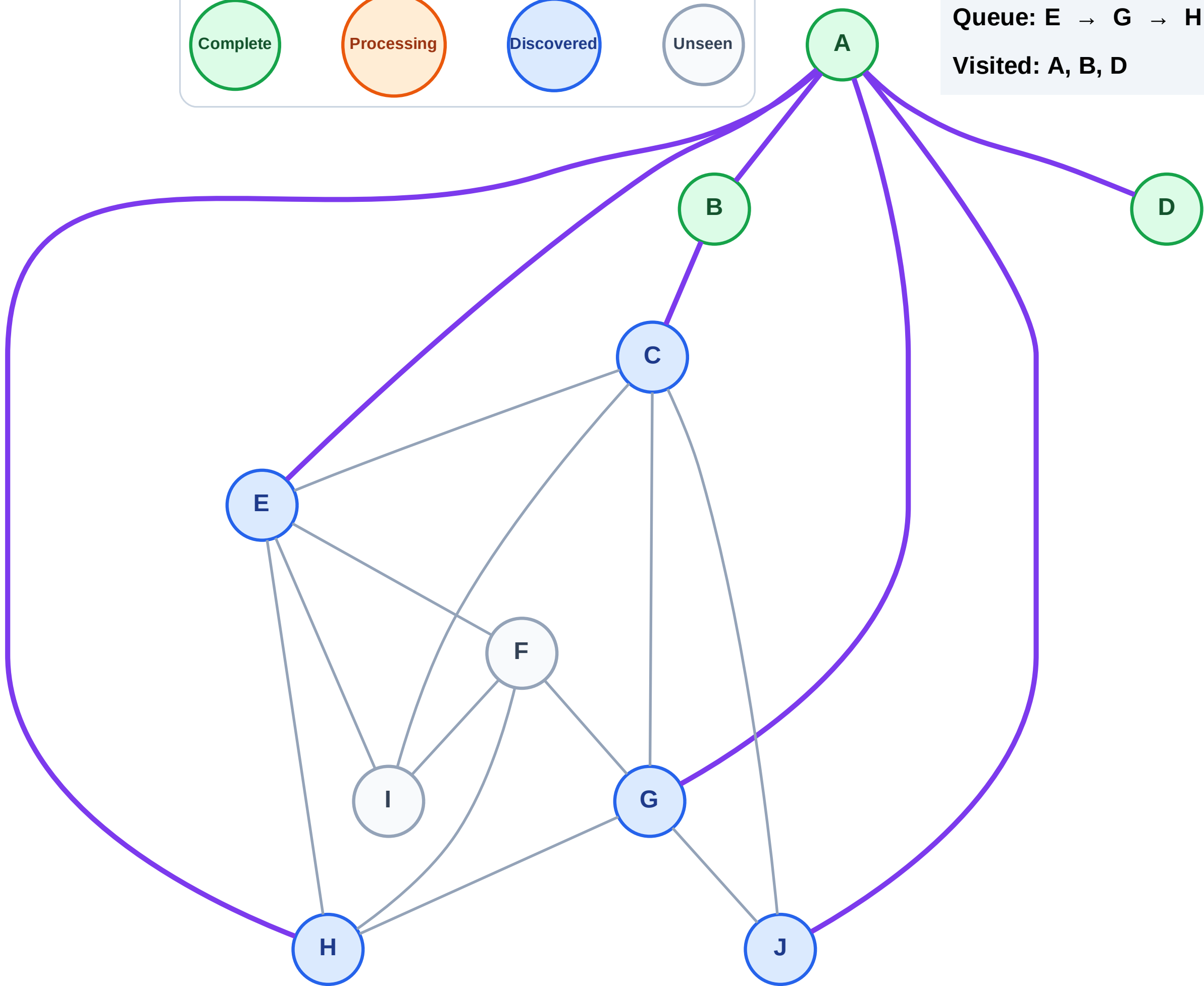
BFS Traversal

Step 7: No new neighbors from D

Legend



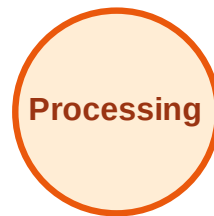
Queue: E → G → H → J → C
Visited: A, B, D



BFS Traversal

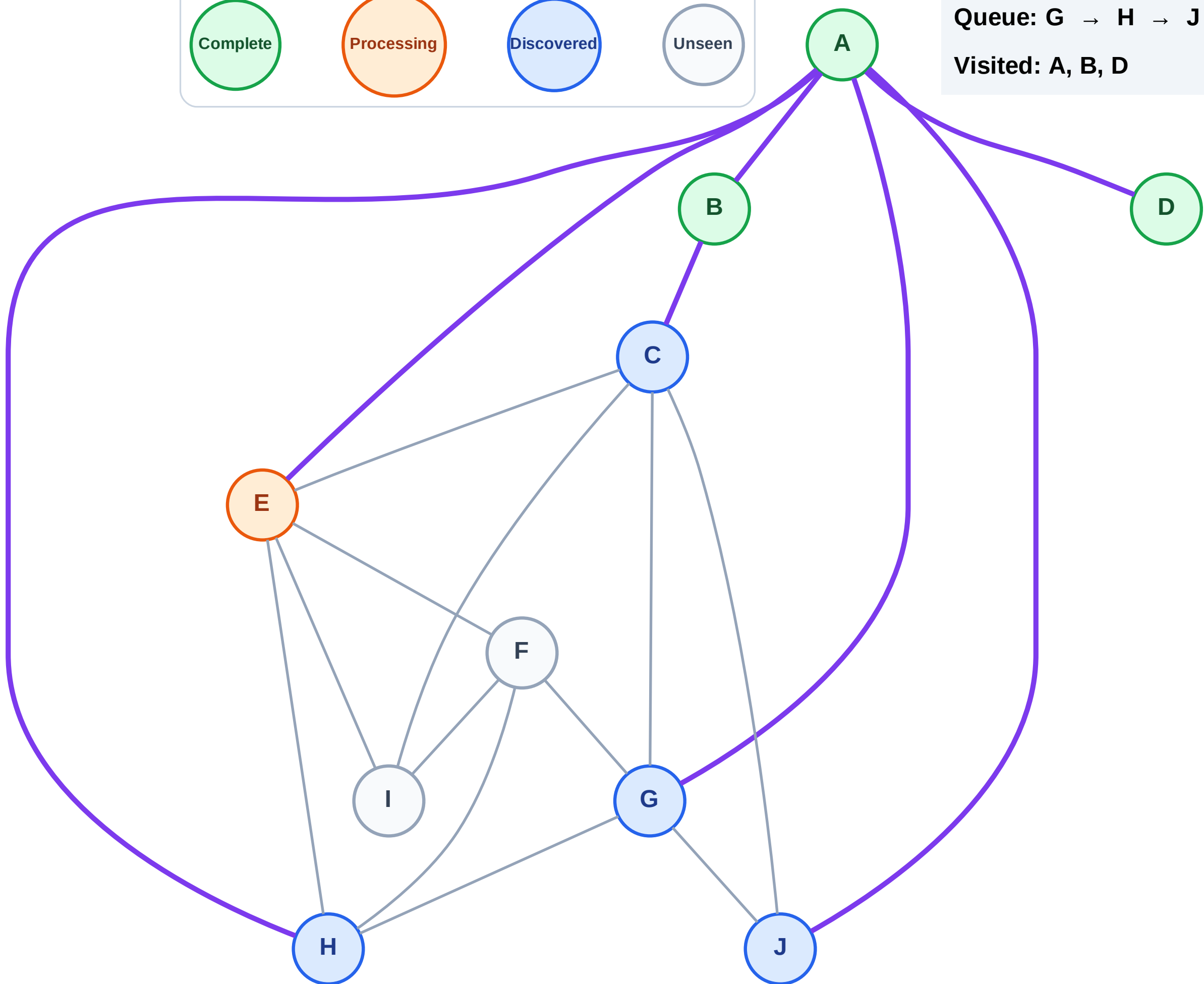
Step 8: Process node E

Legend



Queue: G → H → J → C

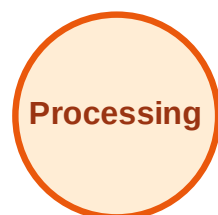
Visited: A, B, D



BFS Traversal

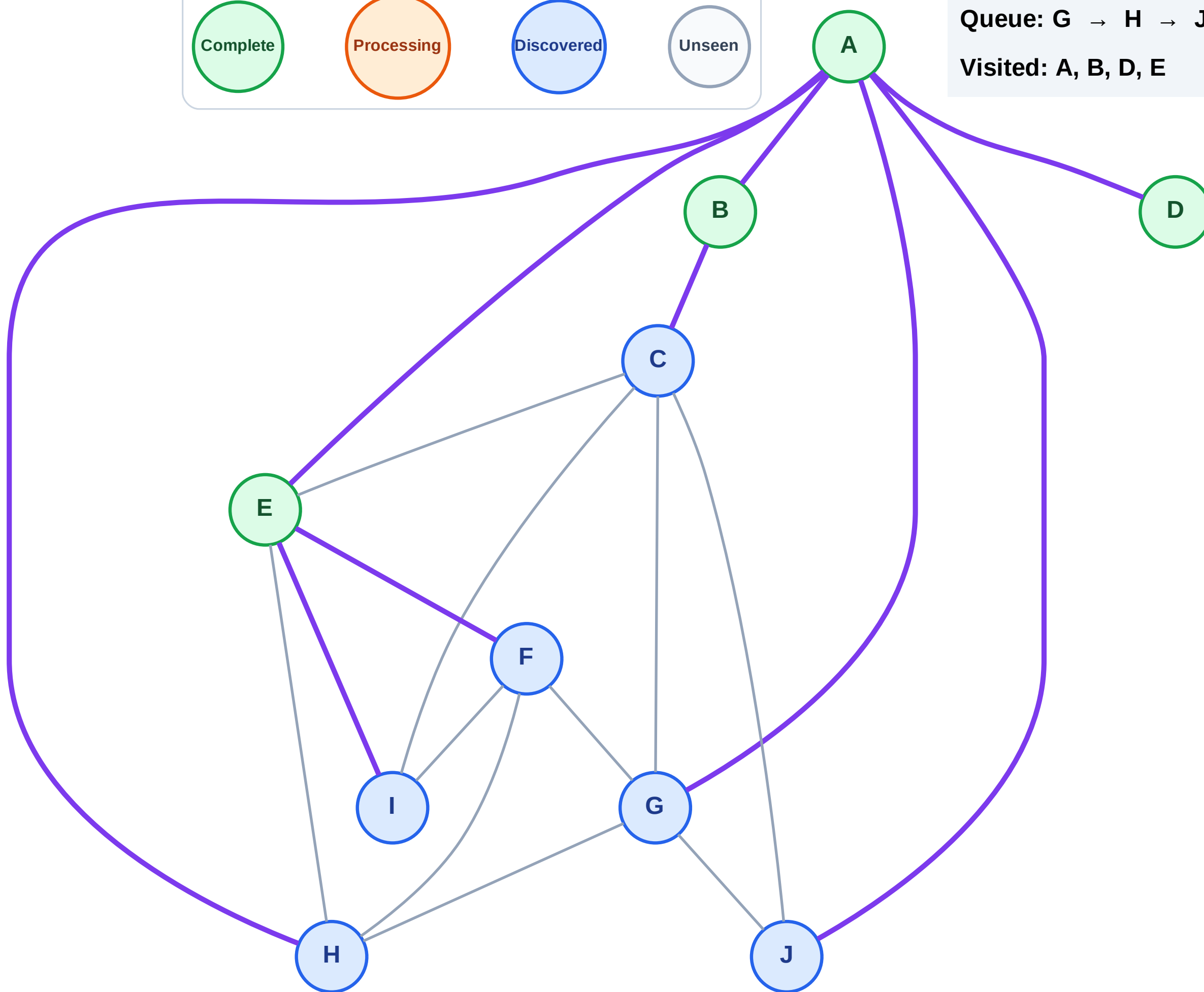
Step 9: Discover F, I from E

Legend



Queue: G → H → J → C → F → I

Visited: A, B, D, E



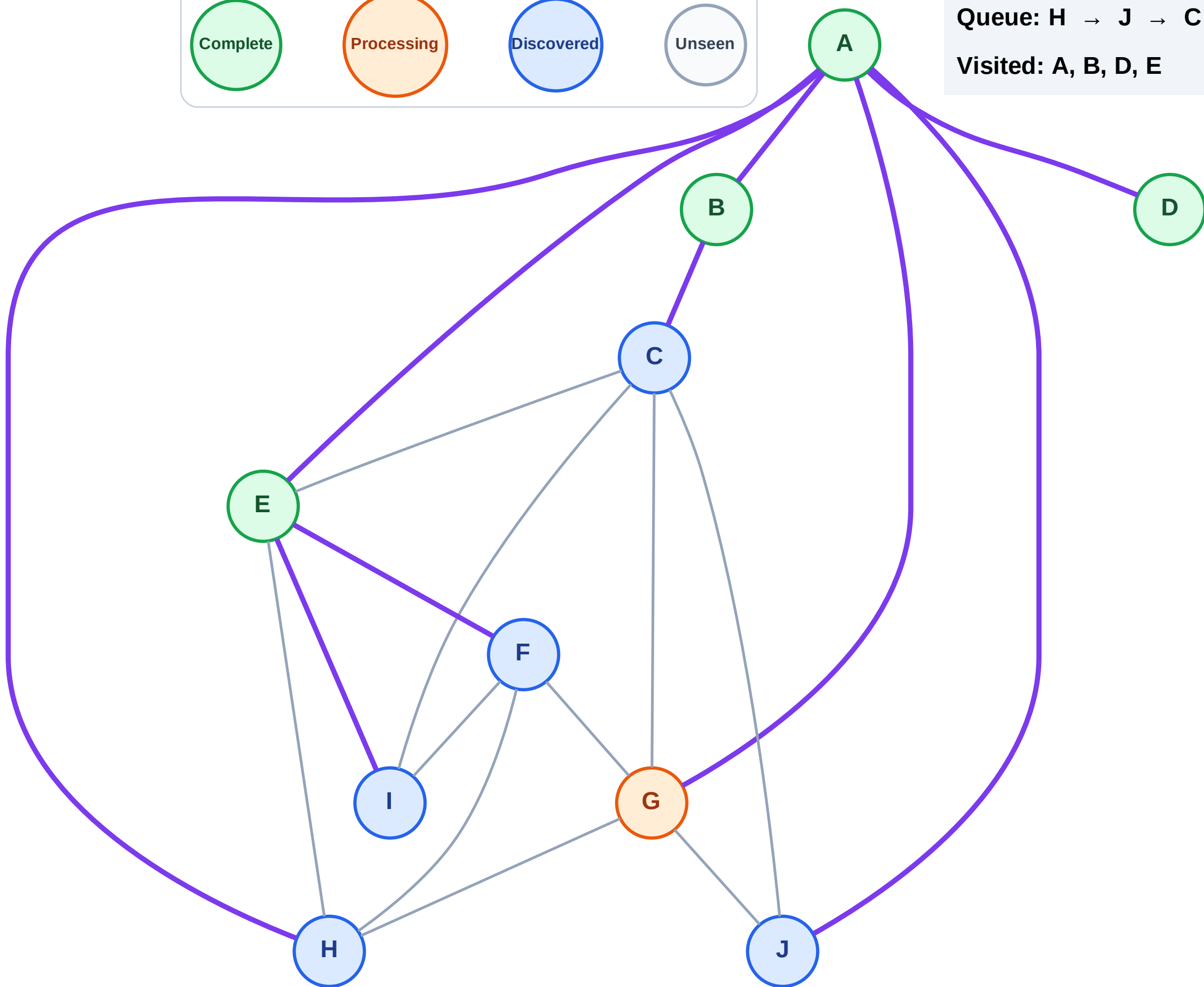
BFS Traversal

Step 10: Process node G

Legend



Queue: H → J → C → F → I
Visited: A, B, D, E



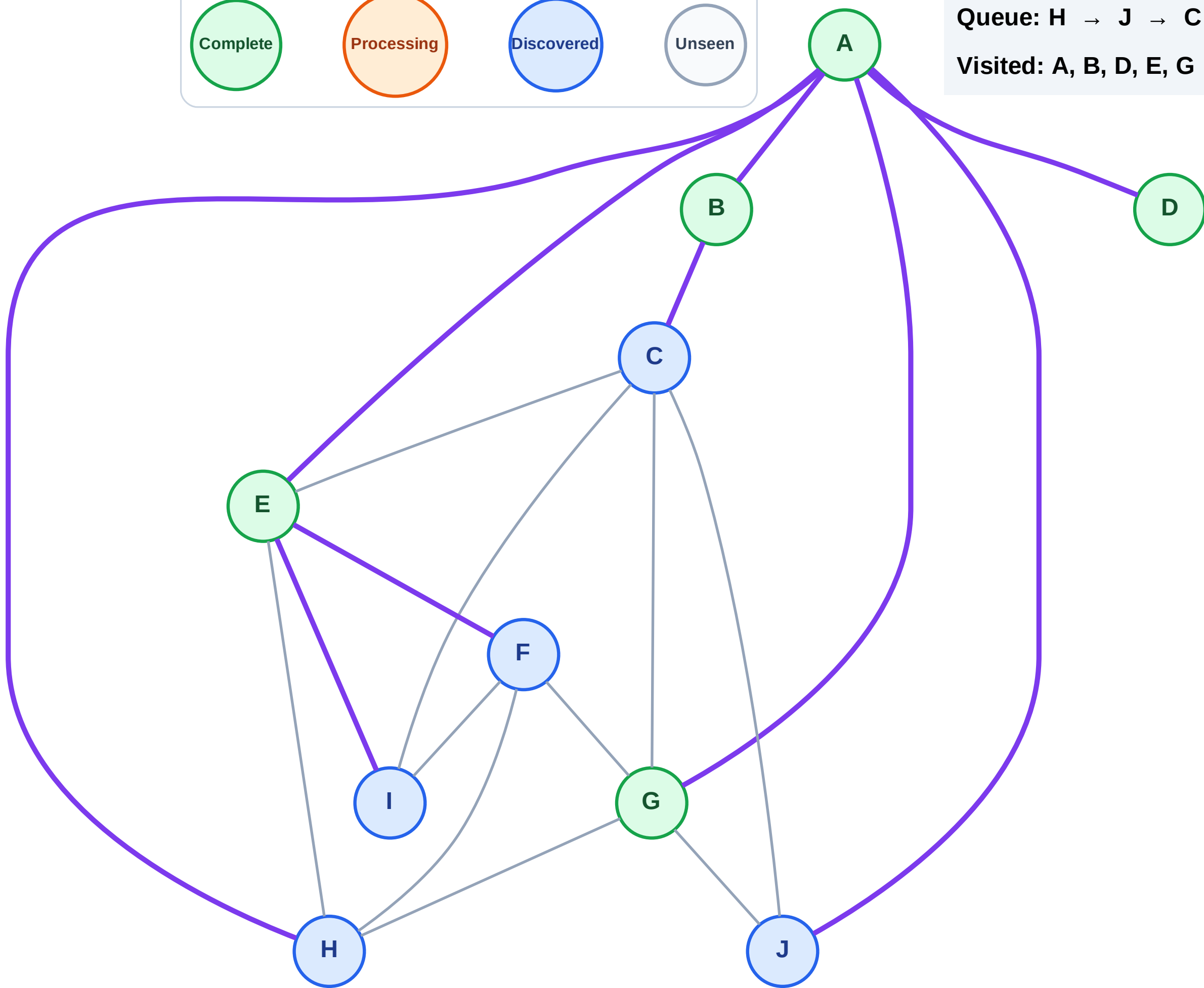
BFS Traversal

Step 11: No new neighbors from G

Legend



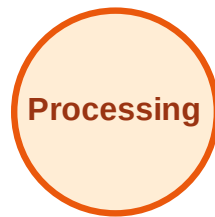
Queue: H → J → C → F → I
Visited: A, B, D, E, G



BFS Traversal

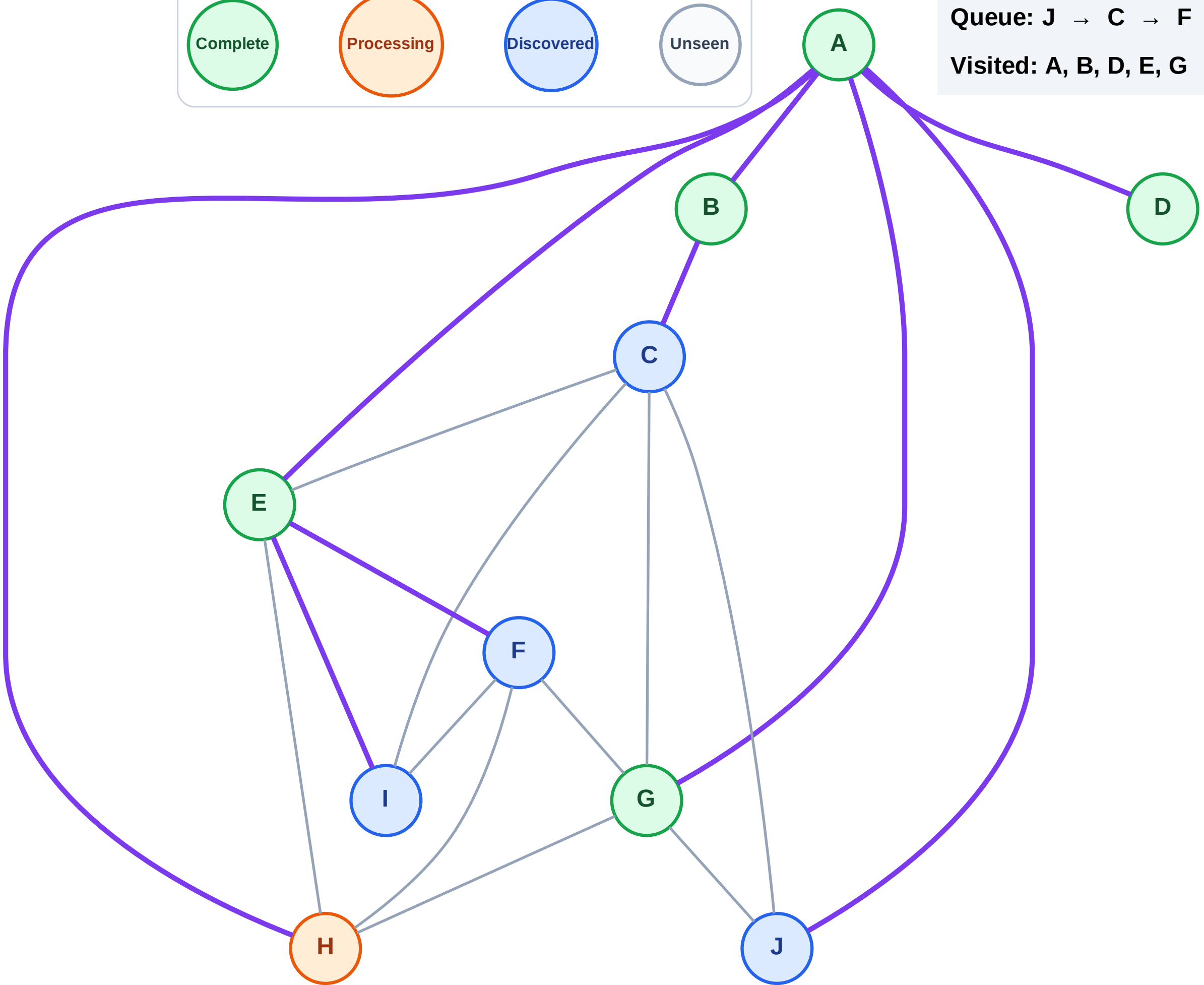
Step 12: Process node H

Legend



Queue: J → C → F → I

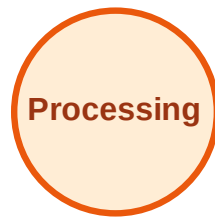
Visited: A, B, D, E, G



BFS Traversal

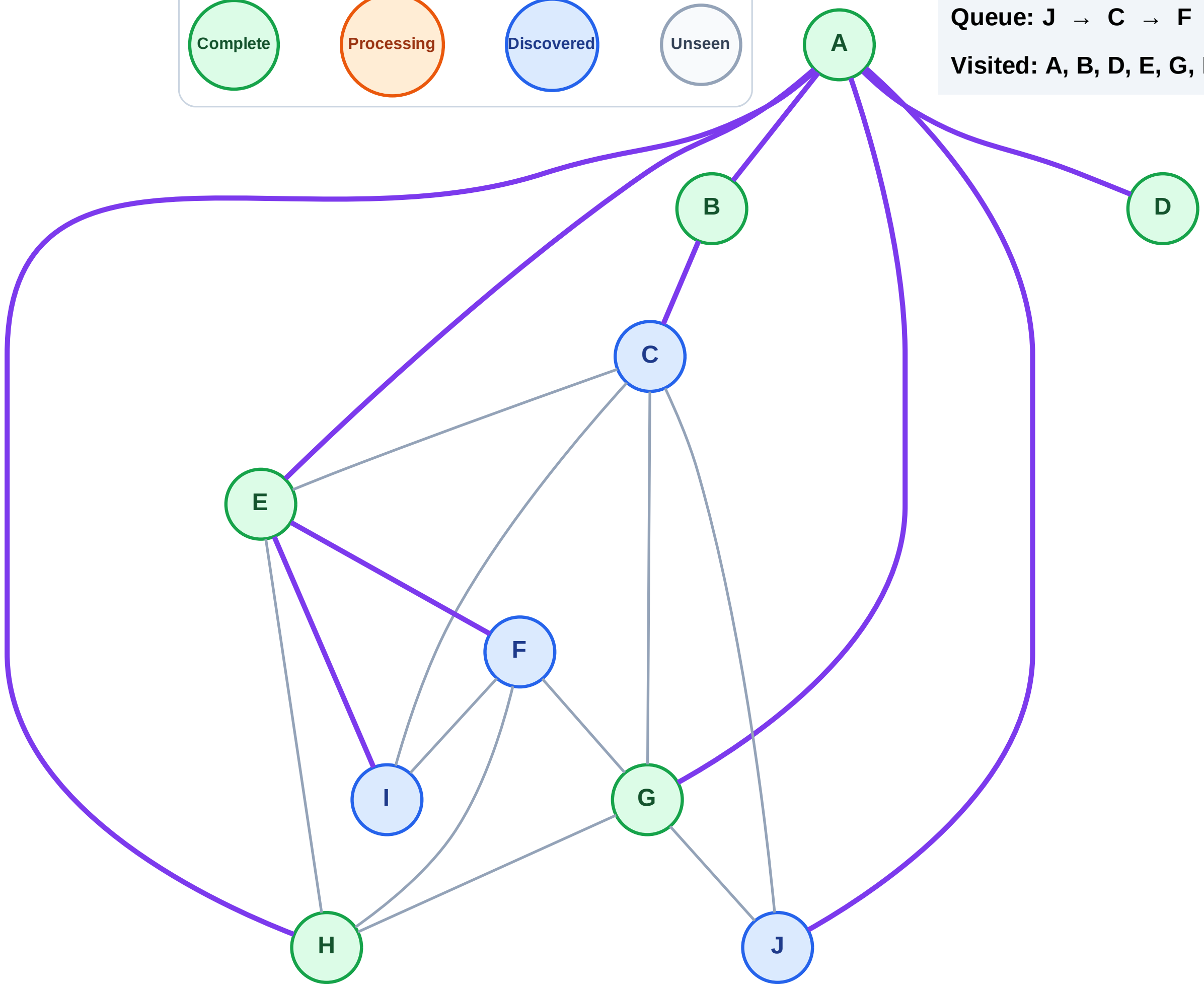
Step 13: No new neighbors from H

Legend



Queue: J → C → F → I

Visited: A, B, D, E, G, H



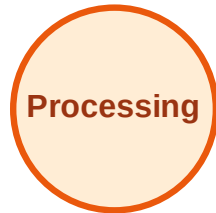
BFS Traversal

Step 14: Process node J

Legend



Complete



Processing

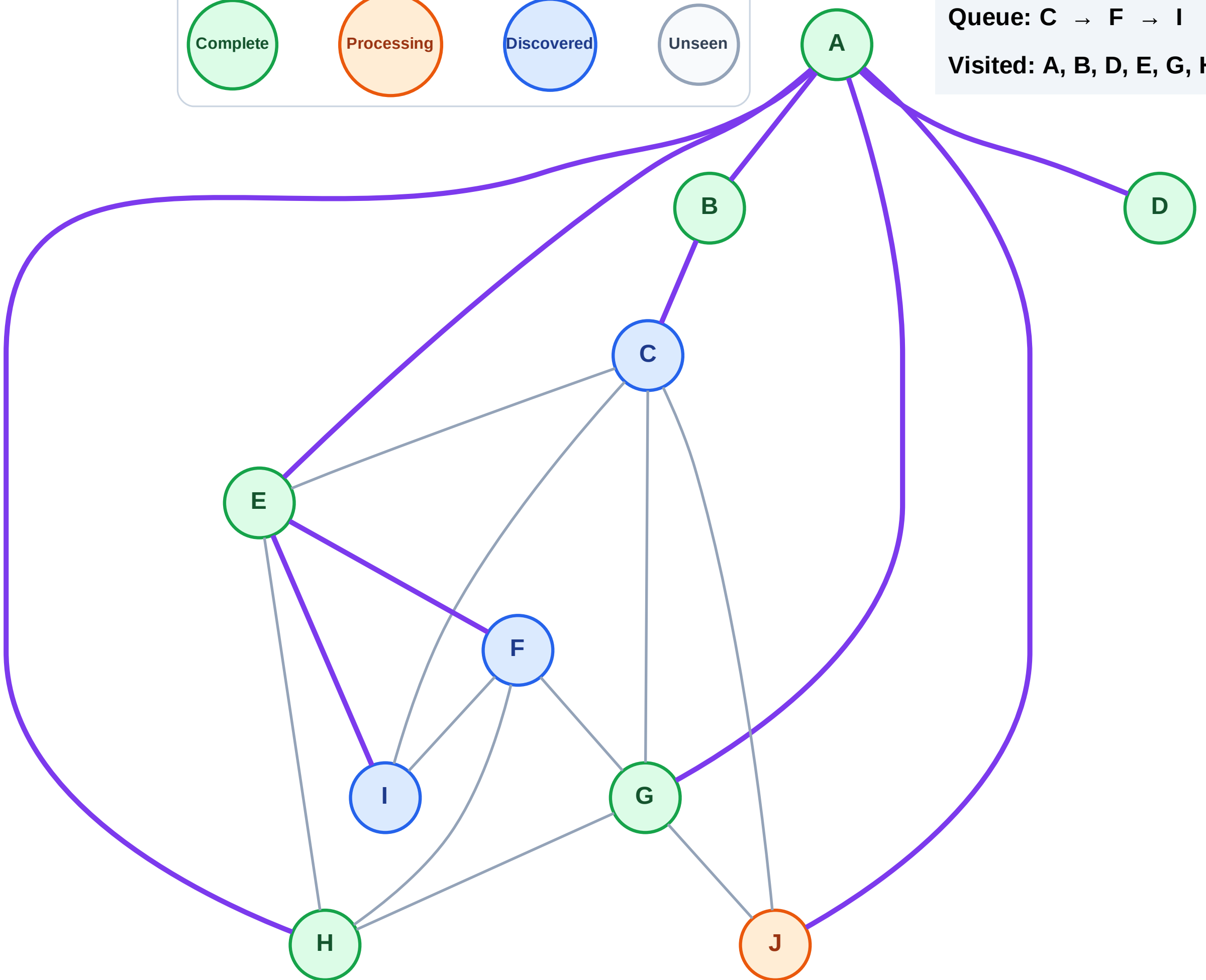


Discovered



Unseen

Queue: C → F → I
Visited: A, B, D, E, G, H



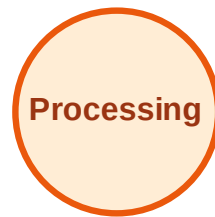
BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing

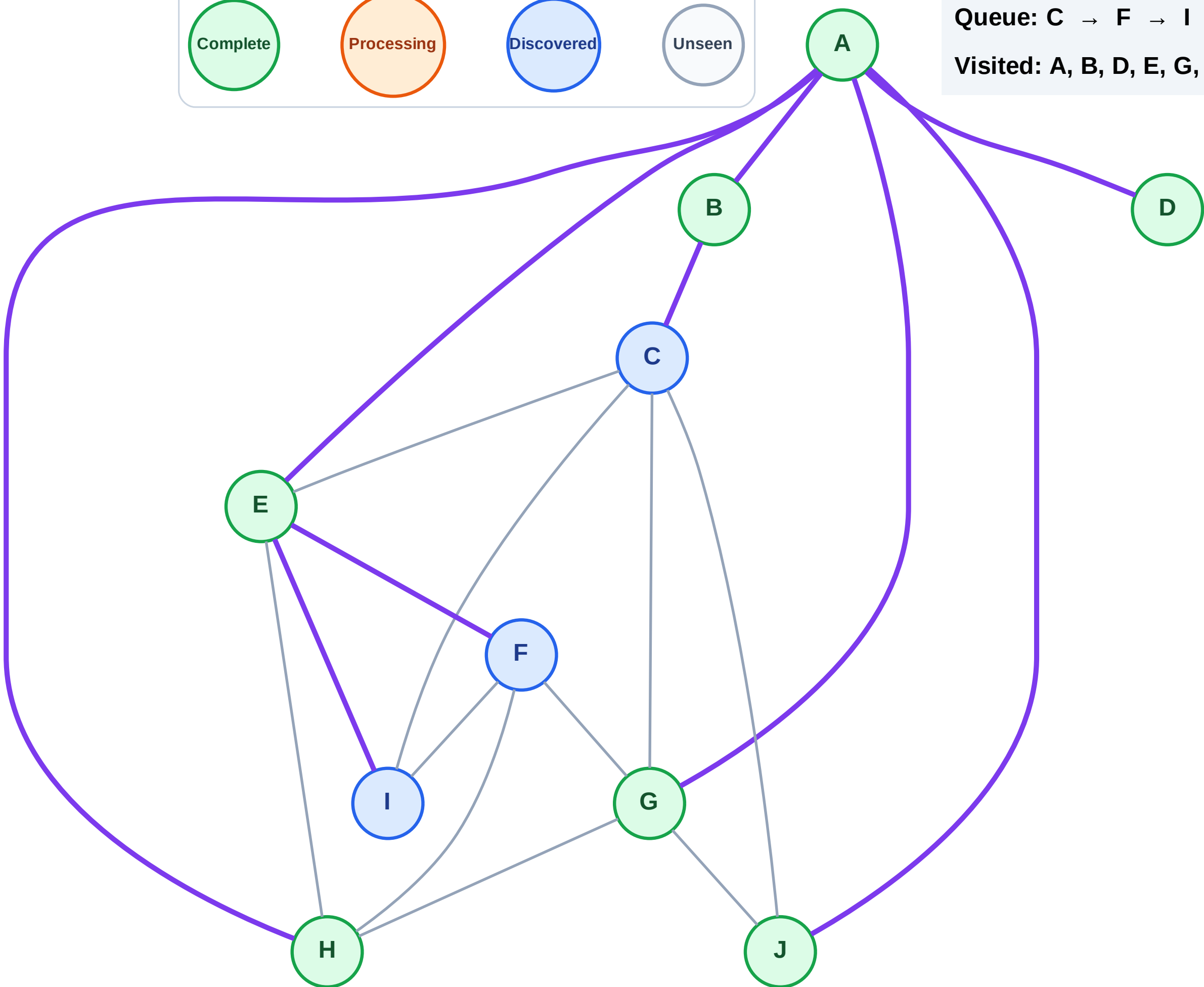


Discovered



Unseen

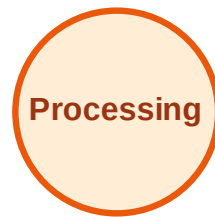
Queue: C → F → I
Visited: A, B, D, E, G, H, J



BFS Traversal

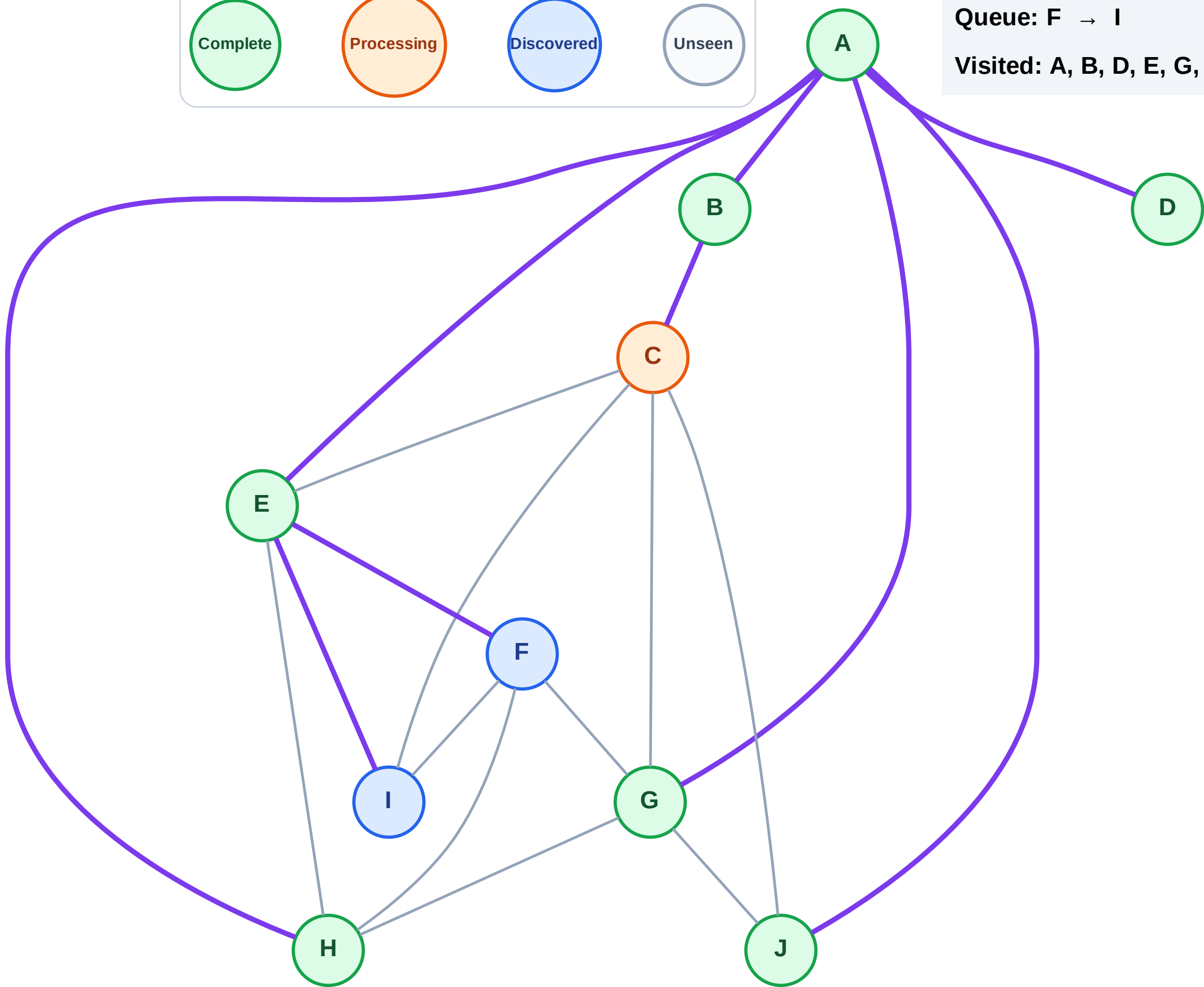
Step 16: Process node C

Legend



Queue: F → I

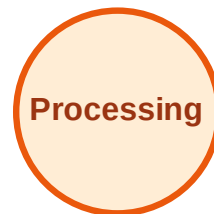
Visited: A, B, D, E, G, H, J



BFS Traversal

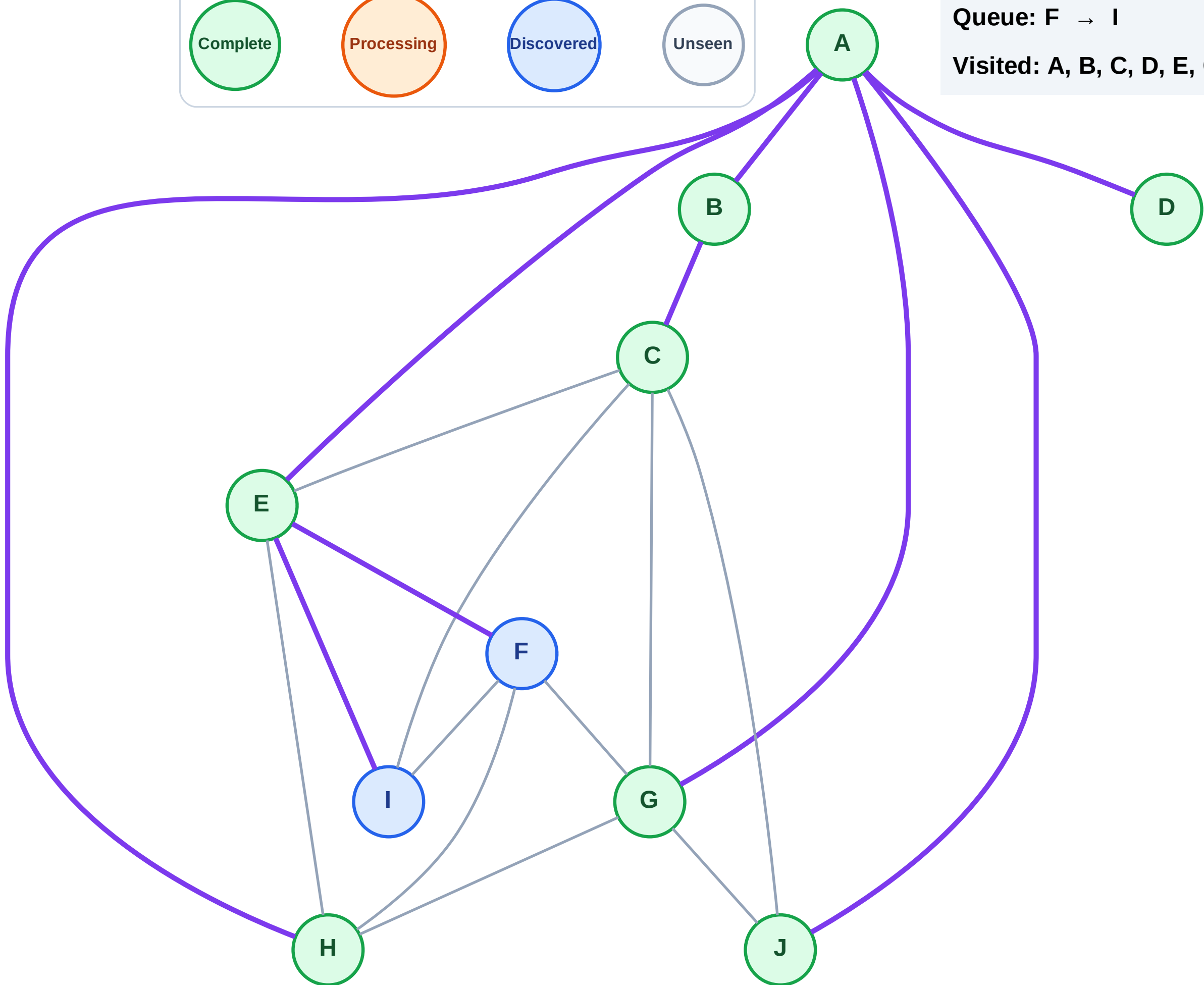
Step 17: No new neighbors from C

Legend



Queue: F → I

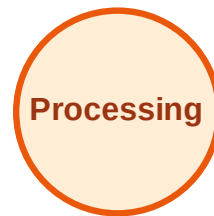
Visited: A, B, C, D, E, G, H, J



BFS Traversal

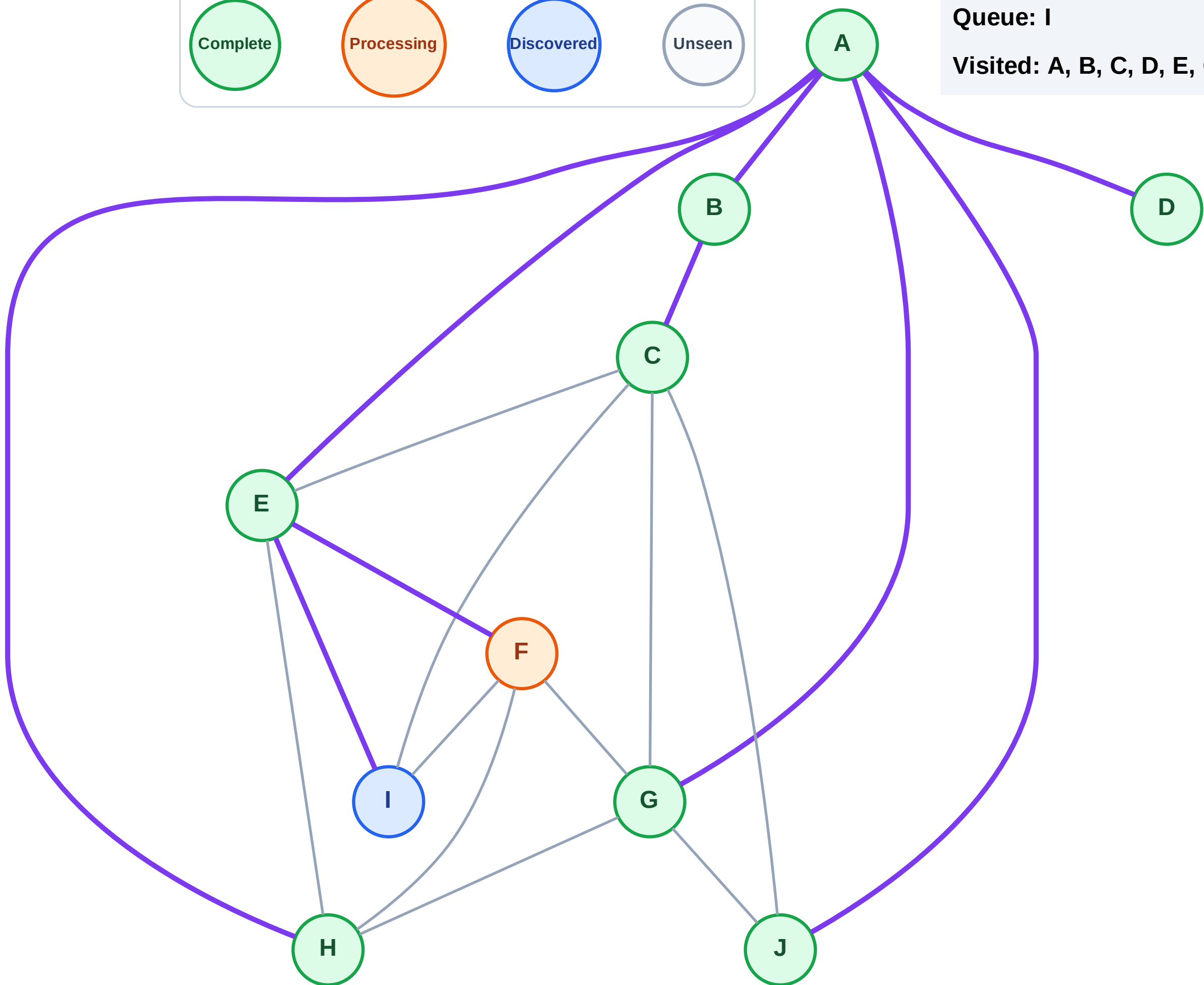
Step 18: Process node F

Legend



Queue: I

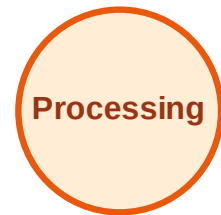
Visited: A, B, C, D, E, G, H, J



BFS Traversal

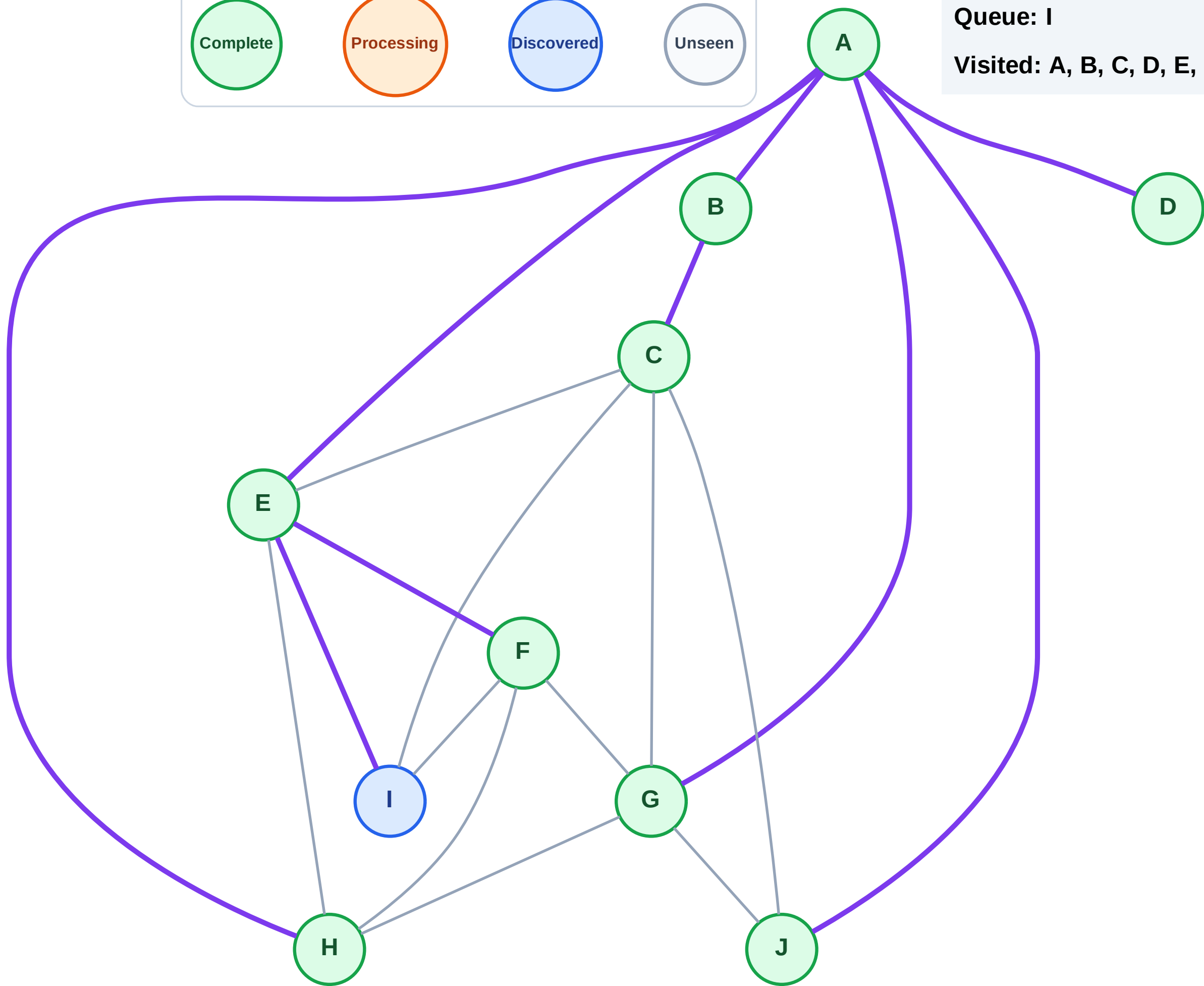
Step 19: No new neighbors from F

Legend



Queue: I

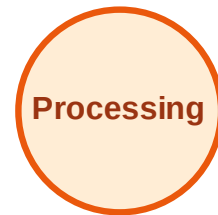
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

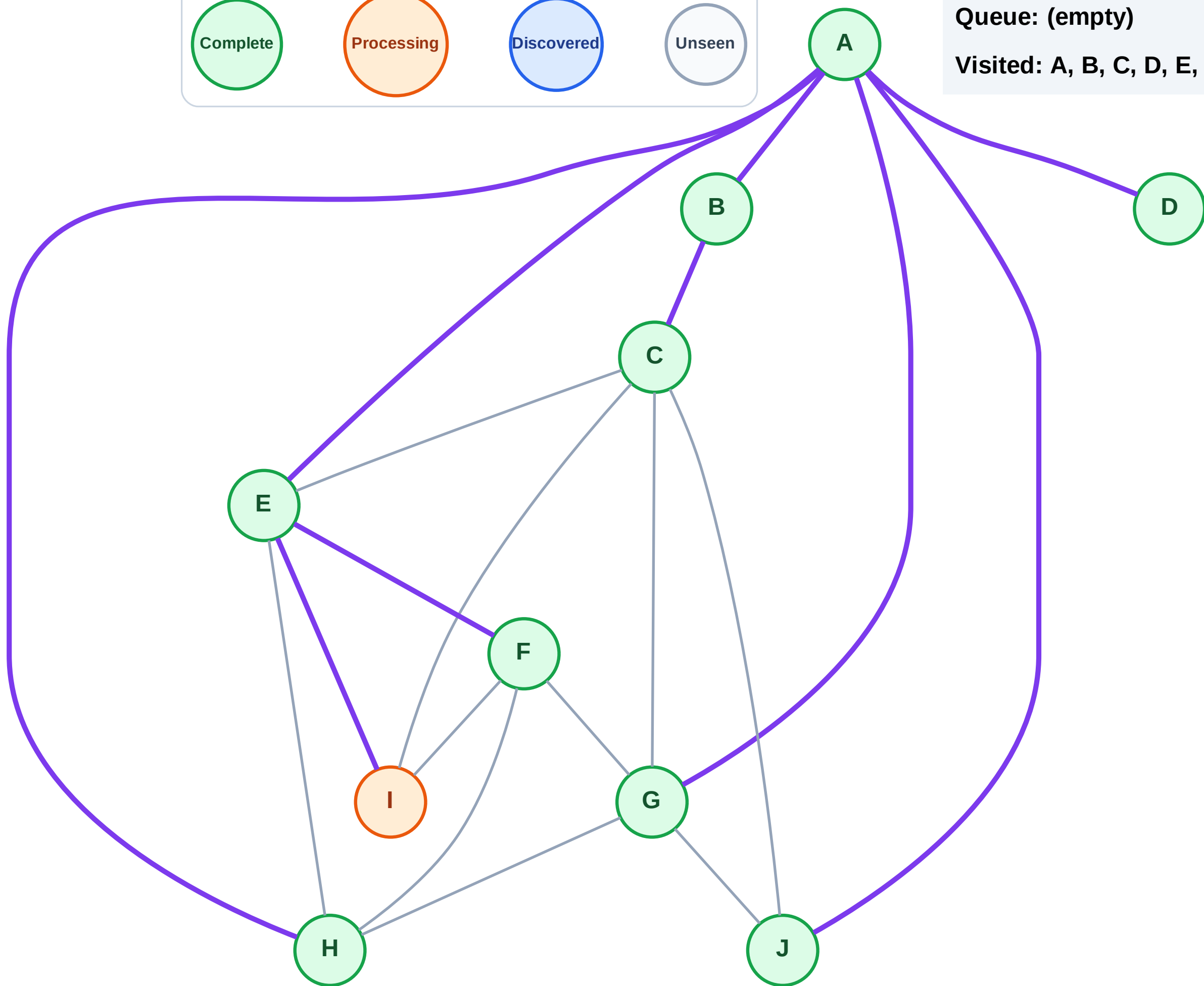
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

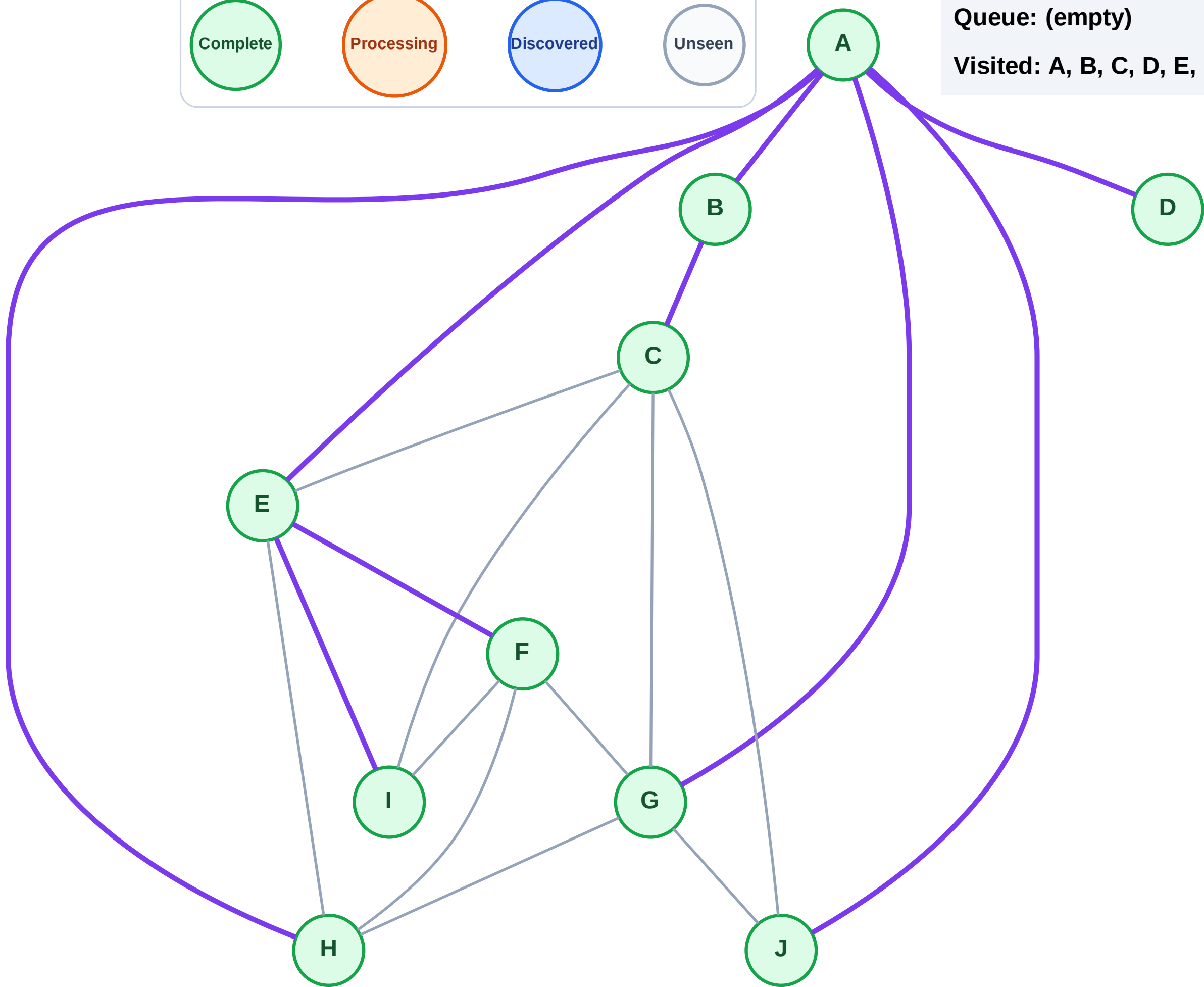
Step 21: No new neighbors from I

Legend



Queue: (empty)

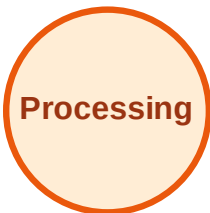
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

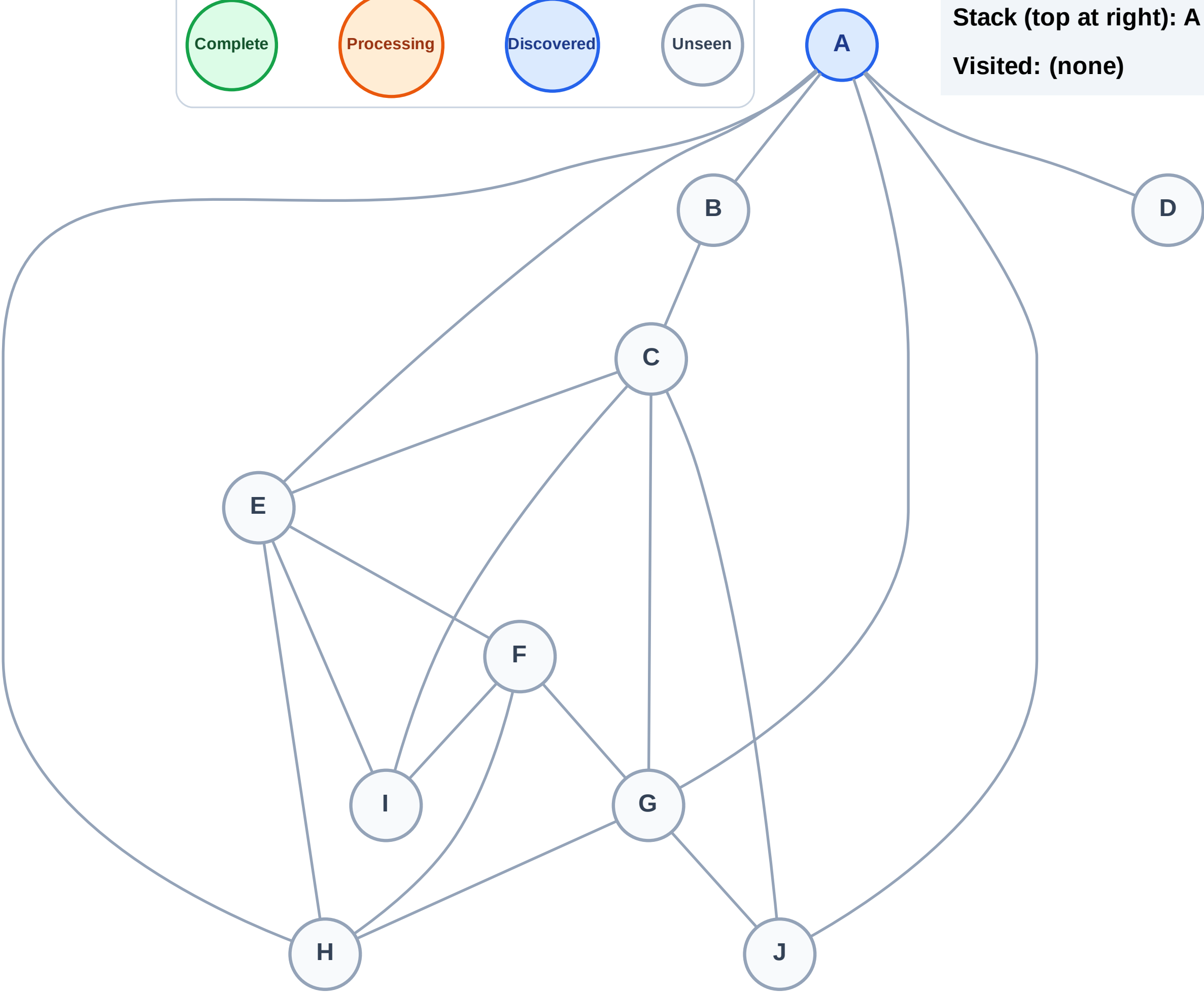
Step 1: Start at A

Legend



Stack (top at right): A

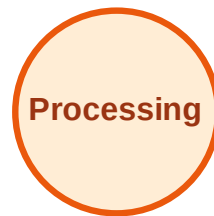
Visited: (none)



DFS Traversal

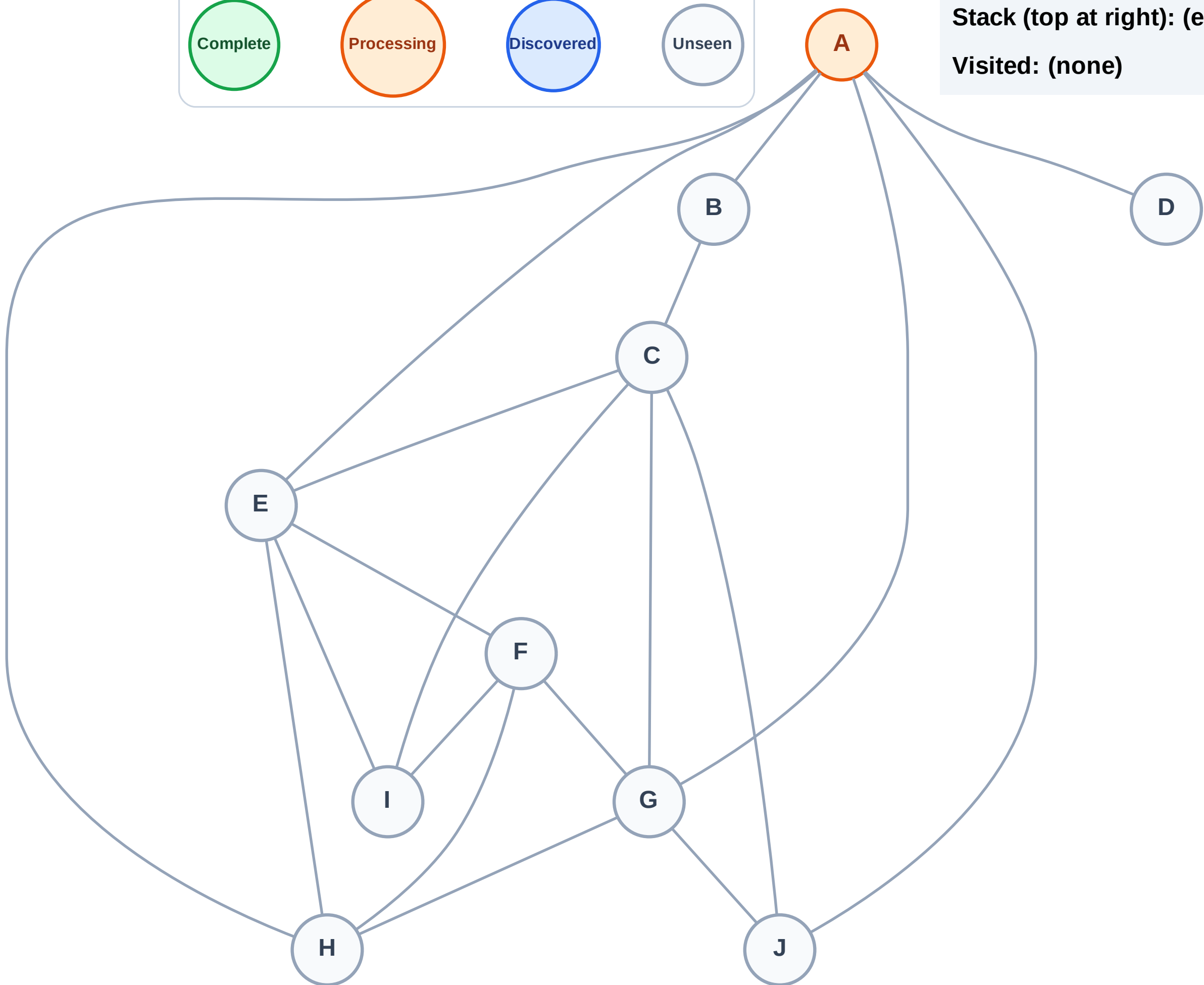
Step 2: Process node A

Legend



Stack (top at right): (empty)

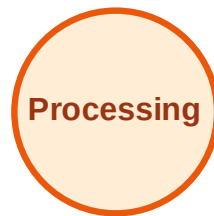
Visited: (none)



DFS Traversal

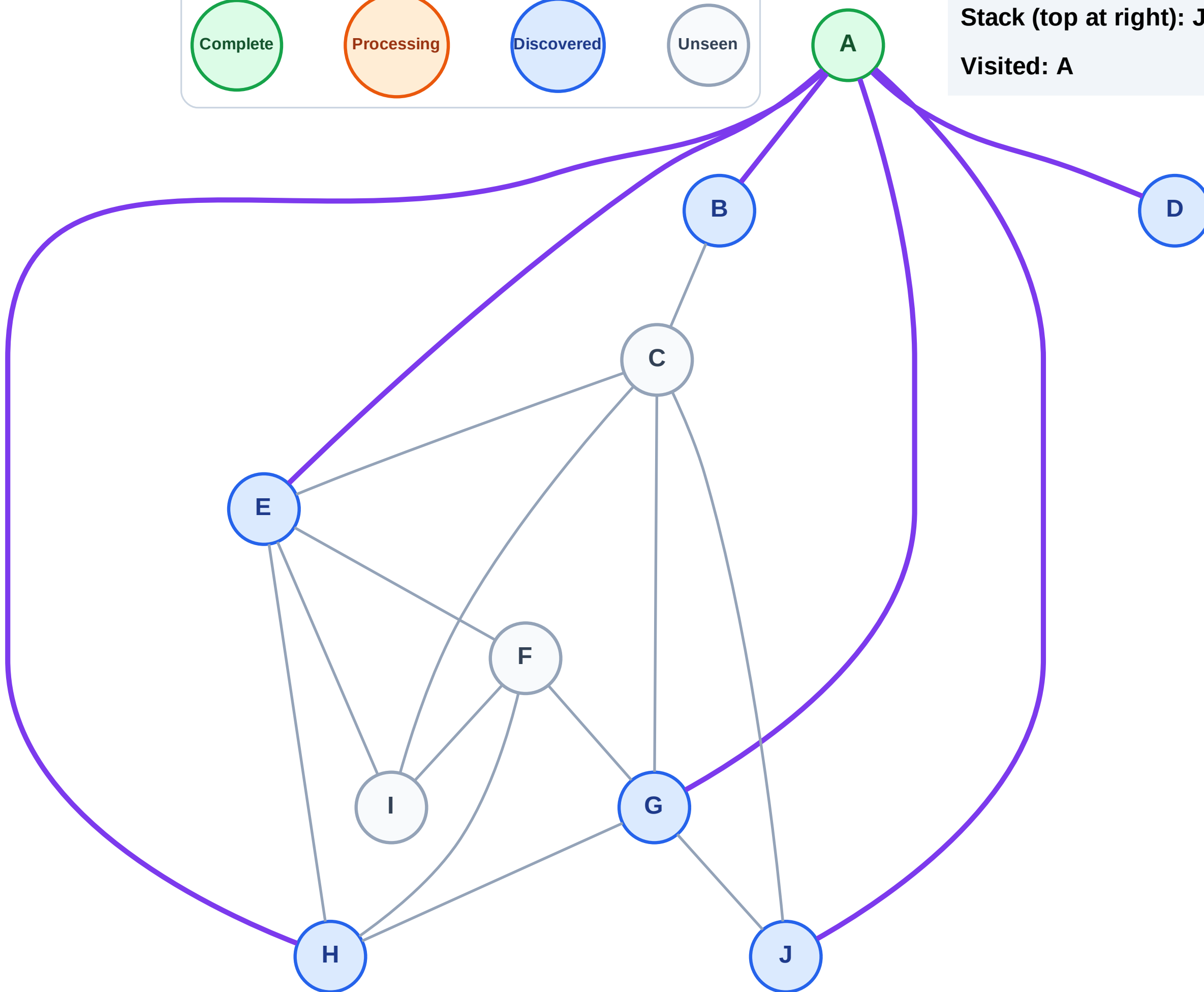
Step 3: Discover J, H, G, E, D, B from A

Legend



Stack (top at right): J | H | G | E | D | B

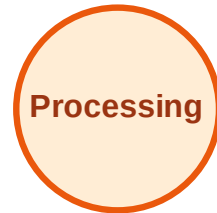
Visited: A



DFS Traversal

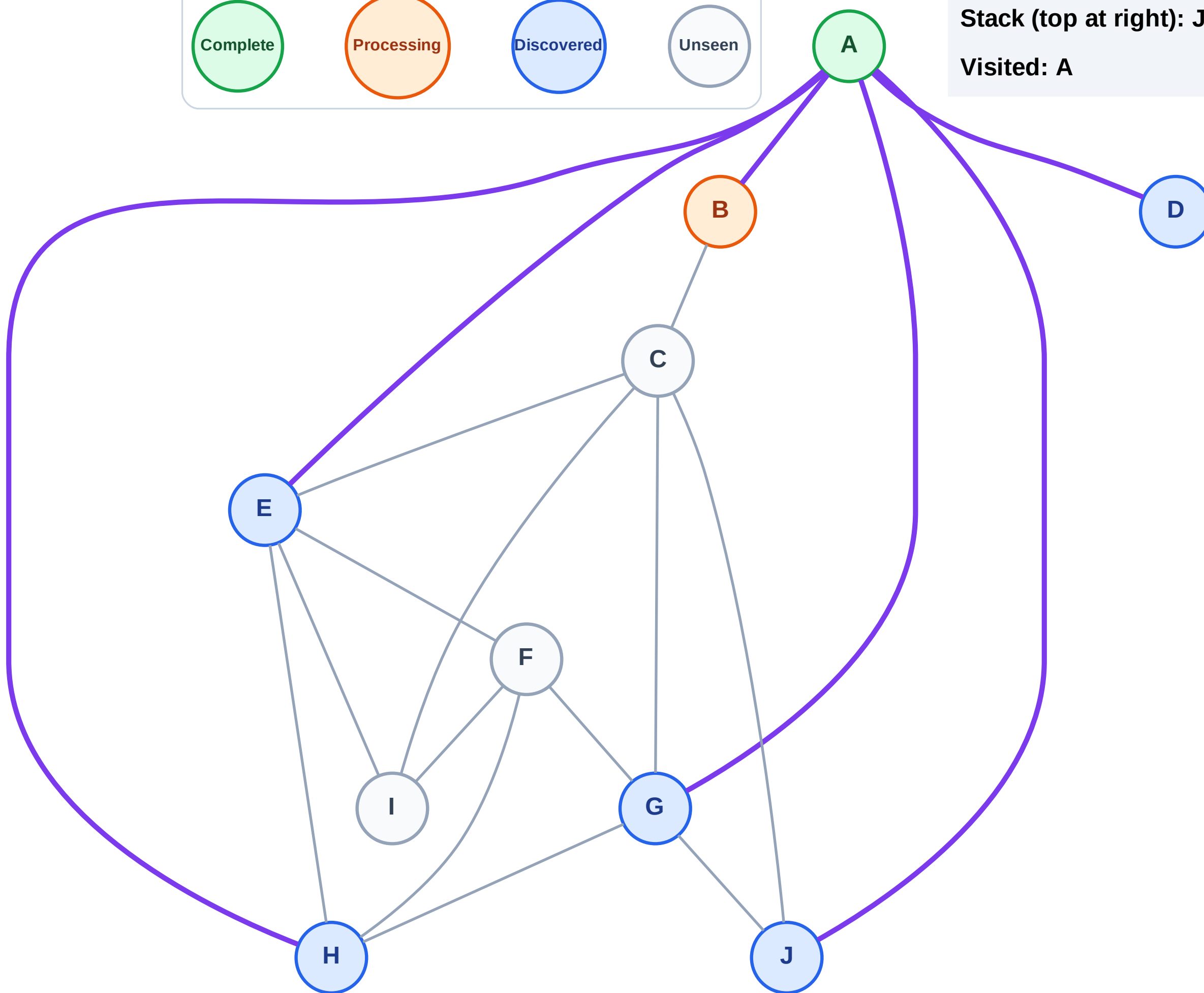
Step 4: Process node B

Legend



Stack (top at right): J | H | G | E | D

Visited: A



Step 5: Discover C from B

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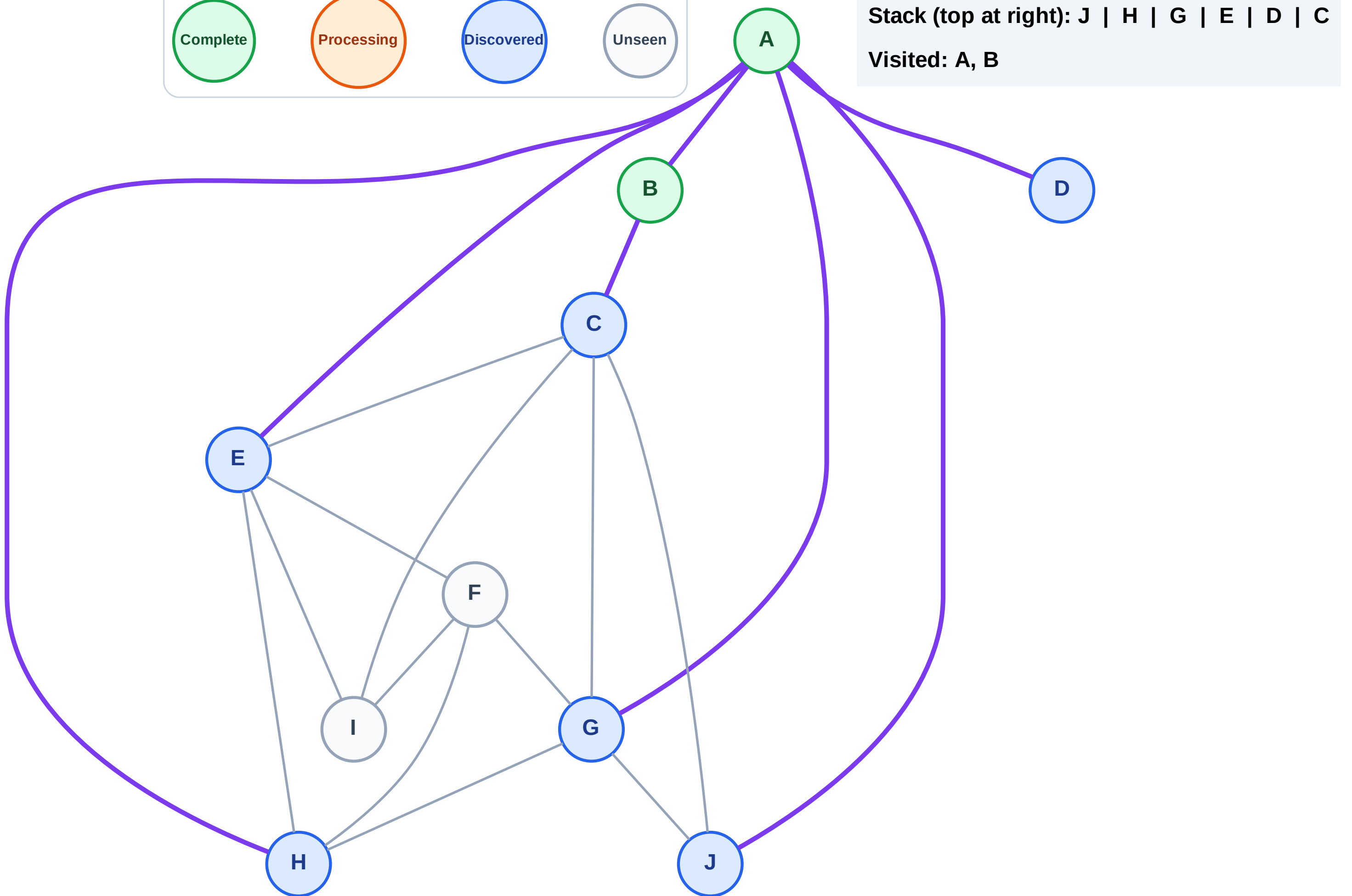
Legend



Complete Processing Discovered Unseen

Unseen

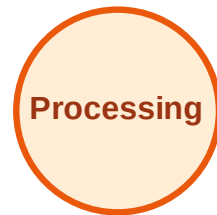
Visited: A, B



DFS Traversal

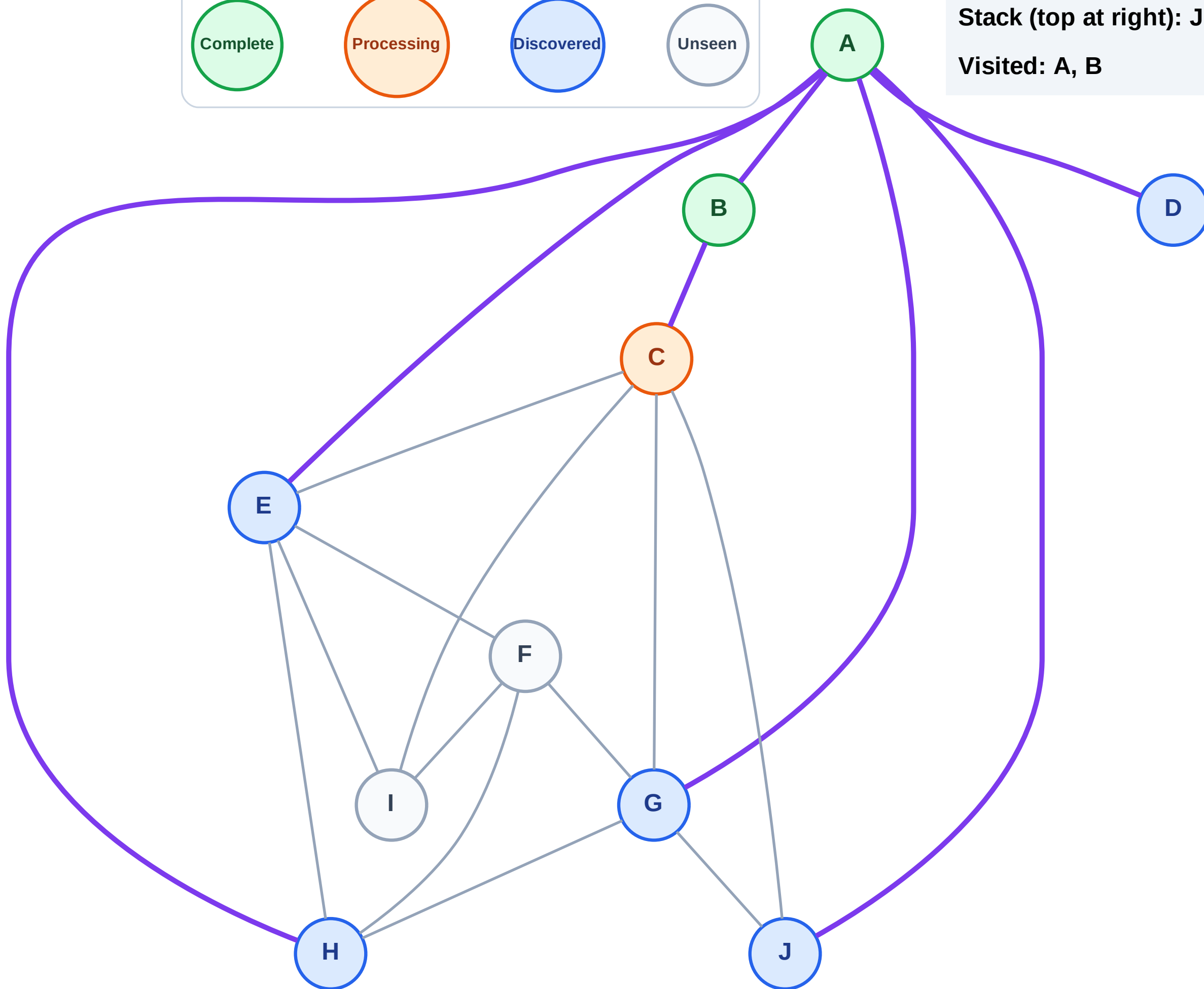
Step 6: Process node C

Legend



Stack (top at right): J | H | G | E | D

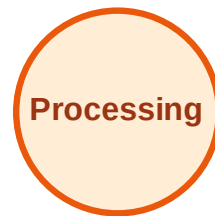
Visited: A, B



DFS Traversal

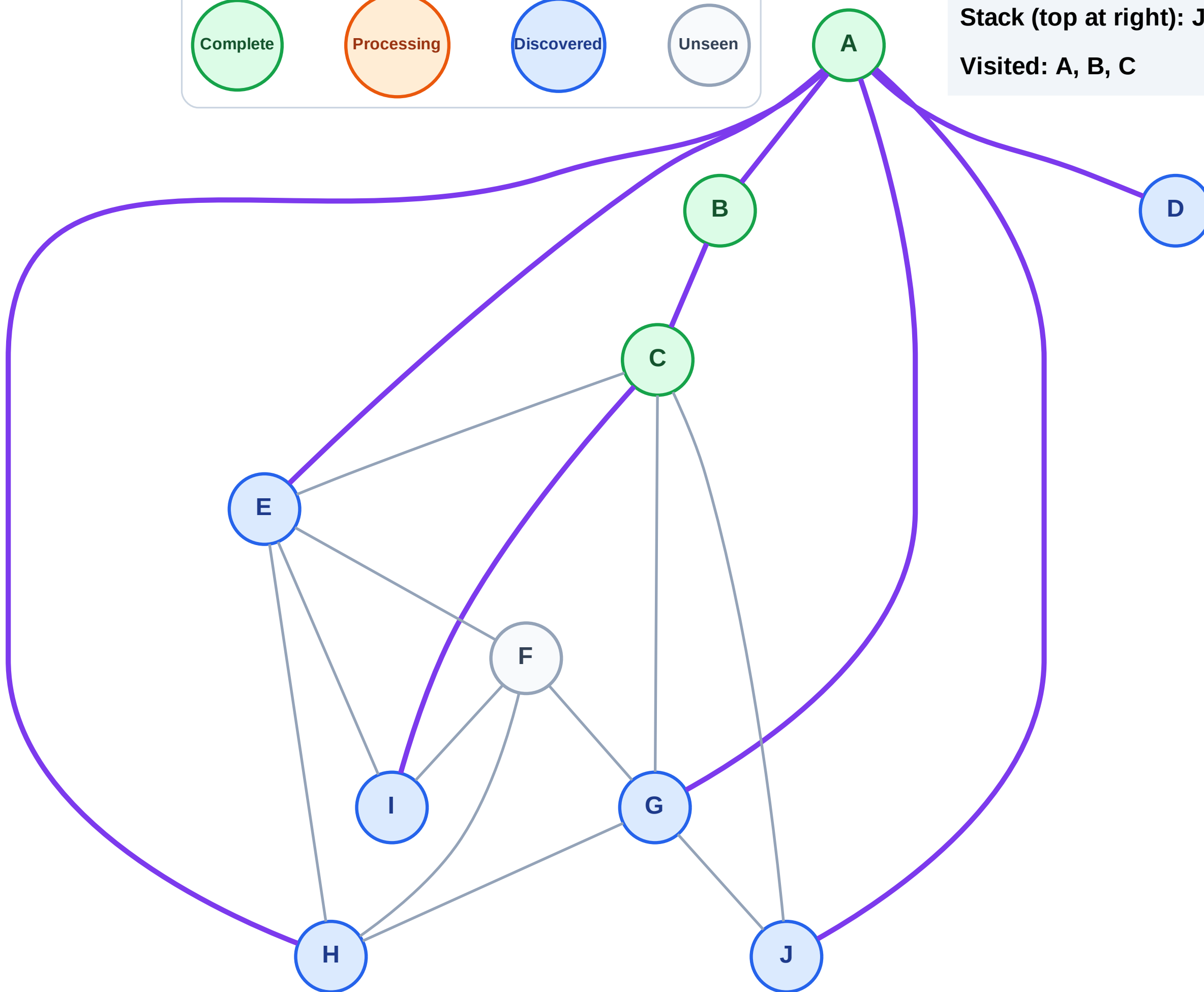
Step 7: Discover I from C

Legend



Stack (top at right): J | H | G | E | D | I

Visited: A, B, C



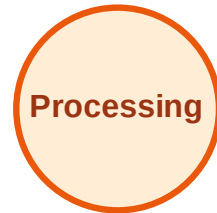
DFS Traversal

Step 8: Process node I

Legend



Complete



Processing



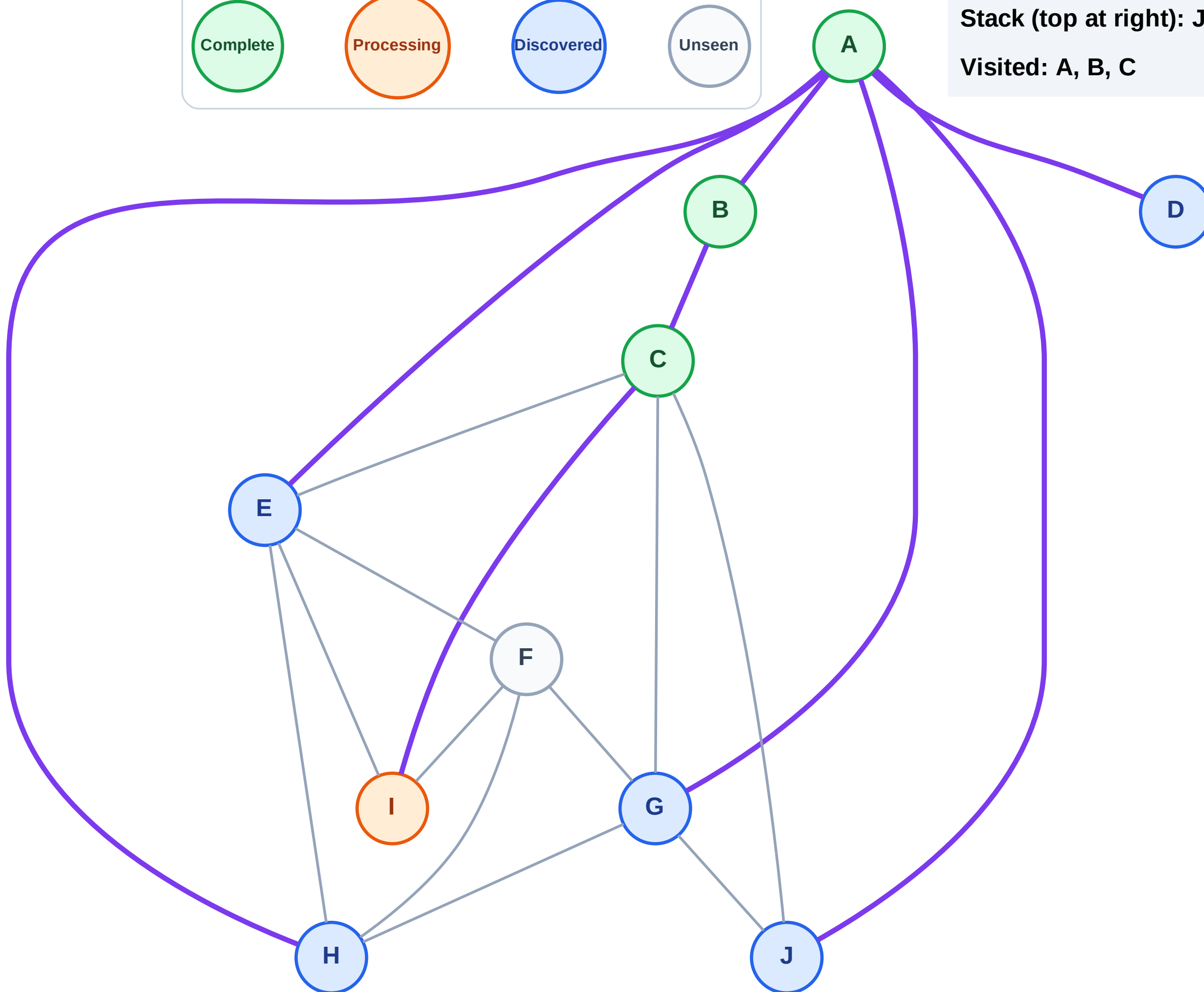
Discovered



Unseen

Stack (top at right): J | H | G | E | D

Visited: A, B, C



DFS Traversal

Step 9: Discover F from I

Legend

Complete

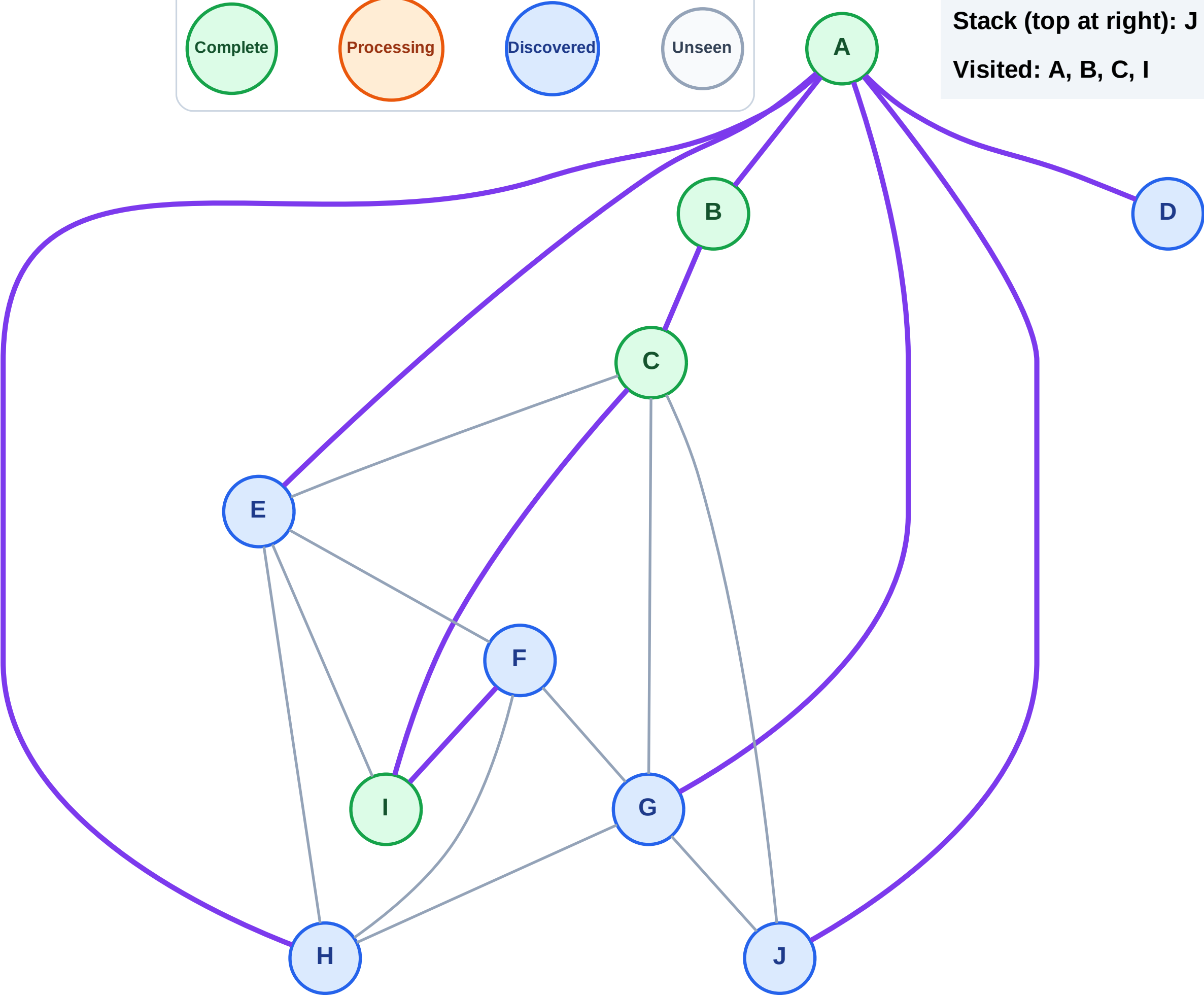
Processing

Discovered

Unseen

Stack (top at right): J | H | G | E | D | F

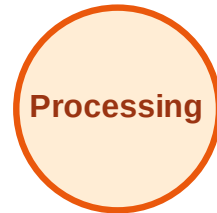
Visited: A, B, C, I



DFS Traversal

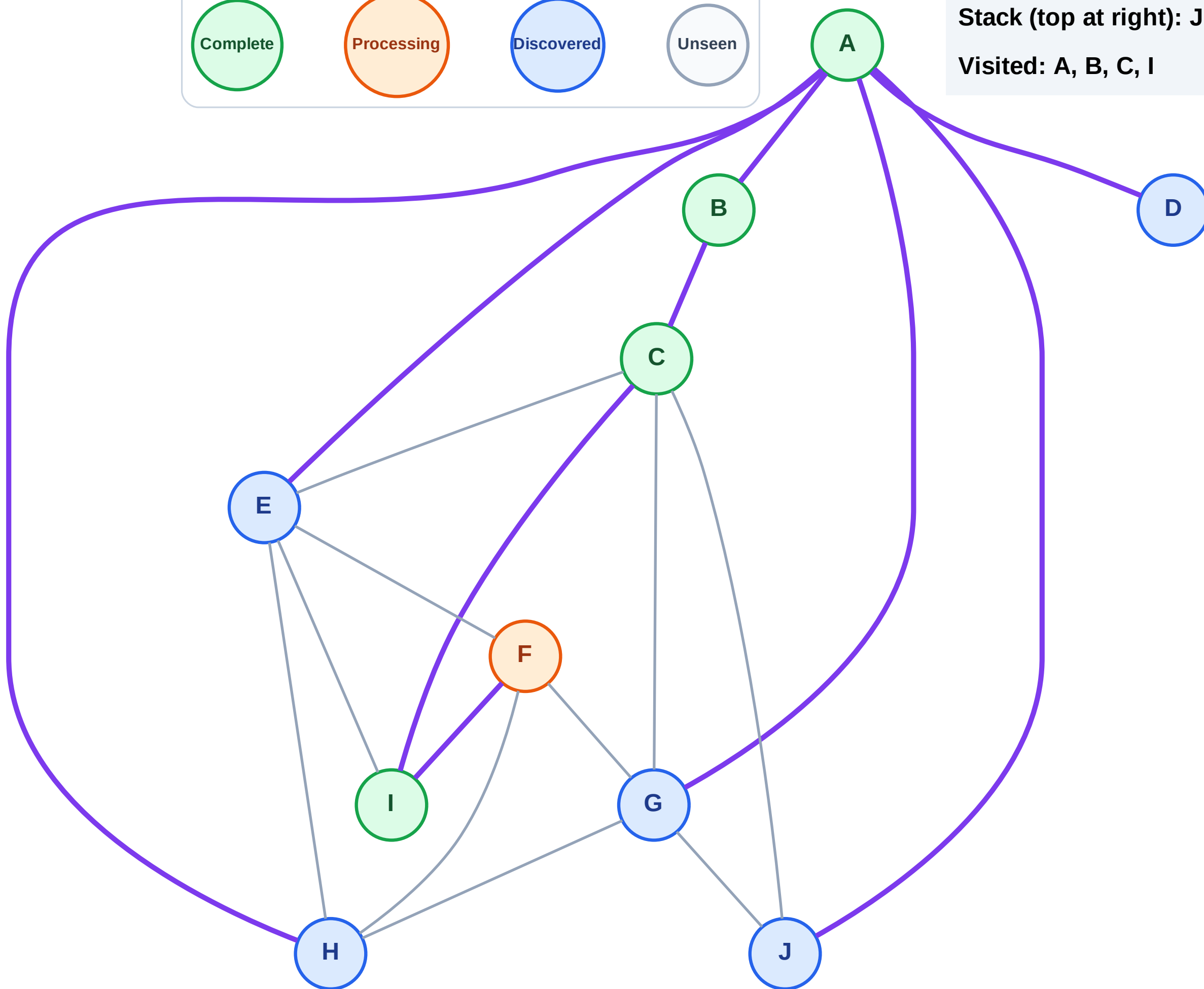
Step 10: Process node F

Legend



Stack (top at right): J | H | G | E | D

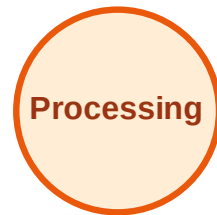
Visited: A, B, C, I



DFS Traversal

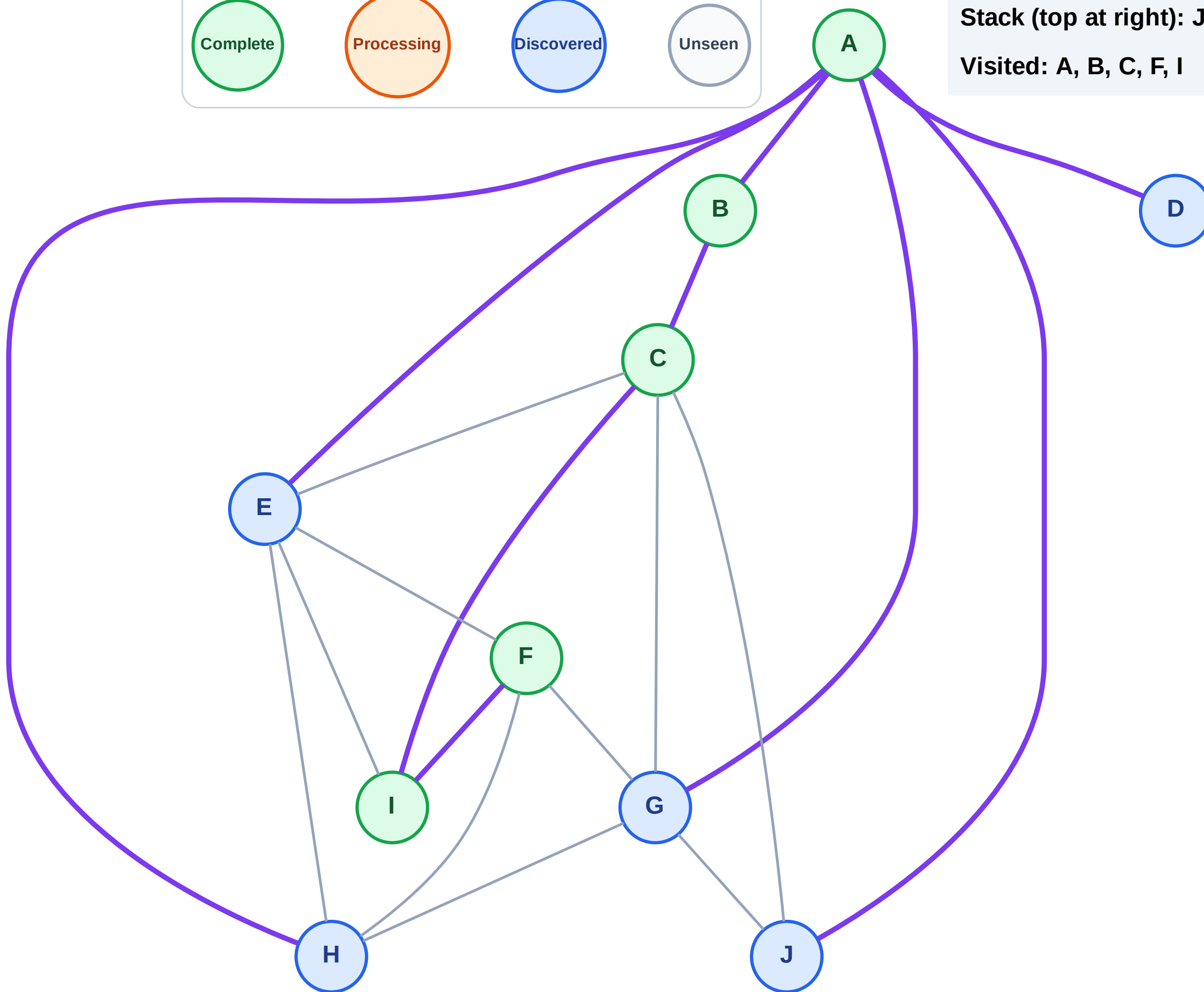
Step 11: No new neighbors from F

Legend



Stack (top at right): J | H | G | E | D

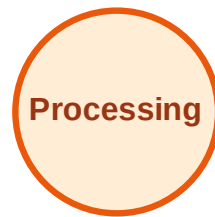
Visited: A, B, C, F, I



DFS Traversal

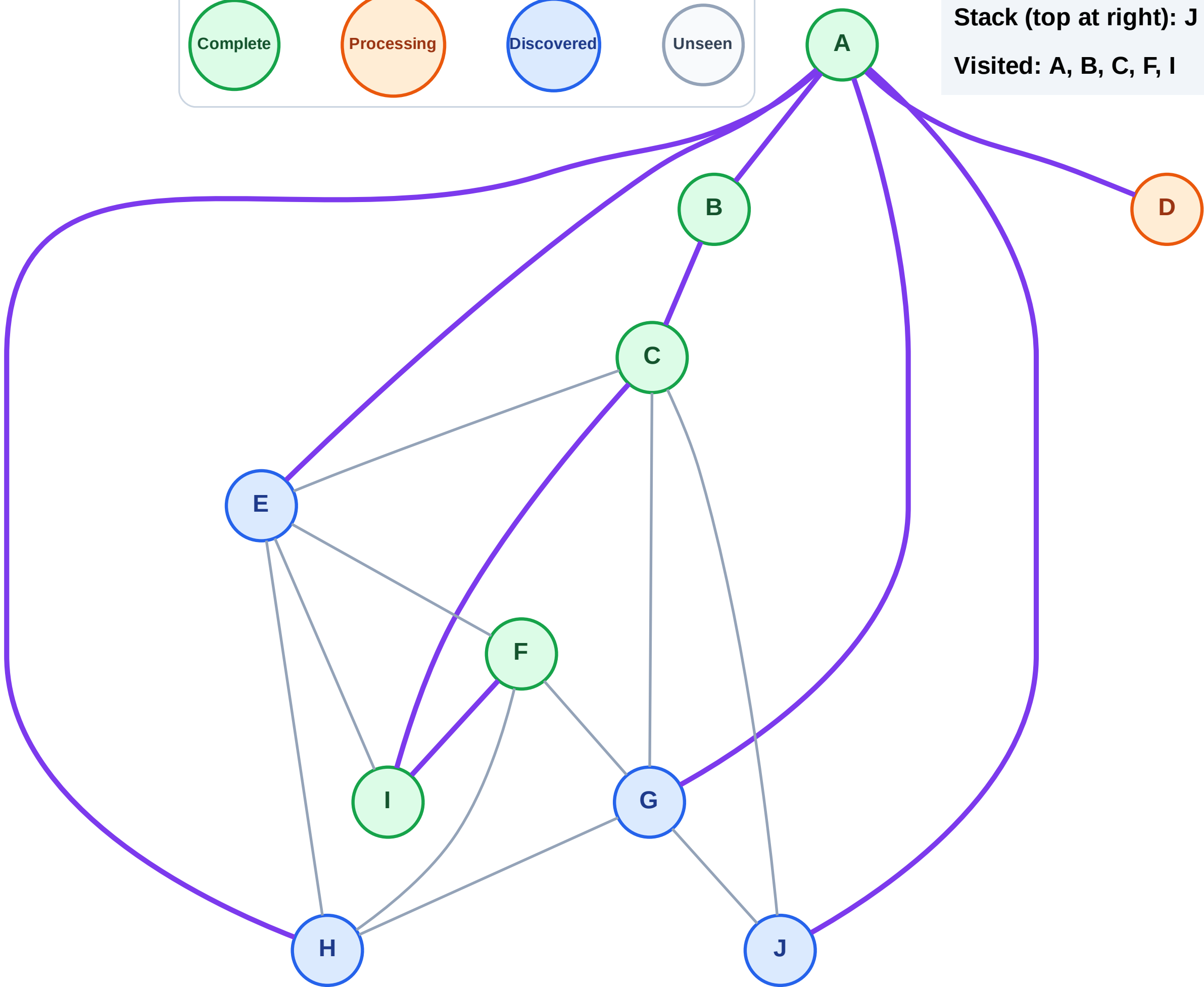
Step 12: Process node D

Legend



Stack (top at right): J | H | G | E

Visited: A, B, C, F, I



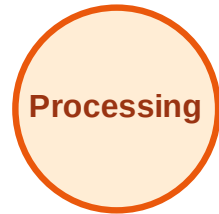
DFS Traversal

Step 13: No new neighbors from D

Legend



Complete



Processing

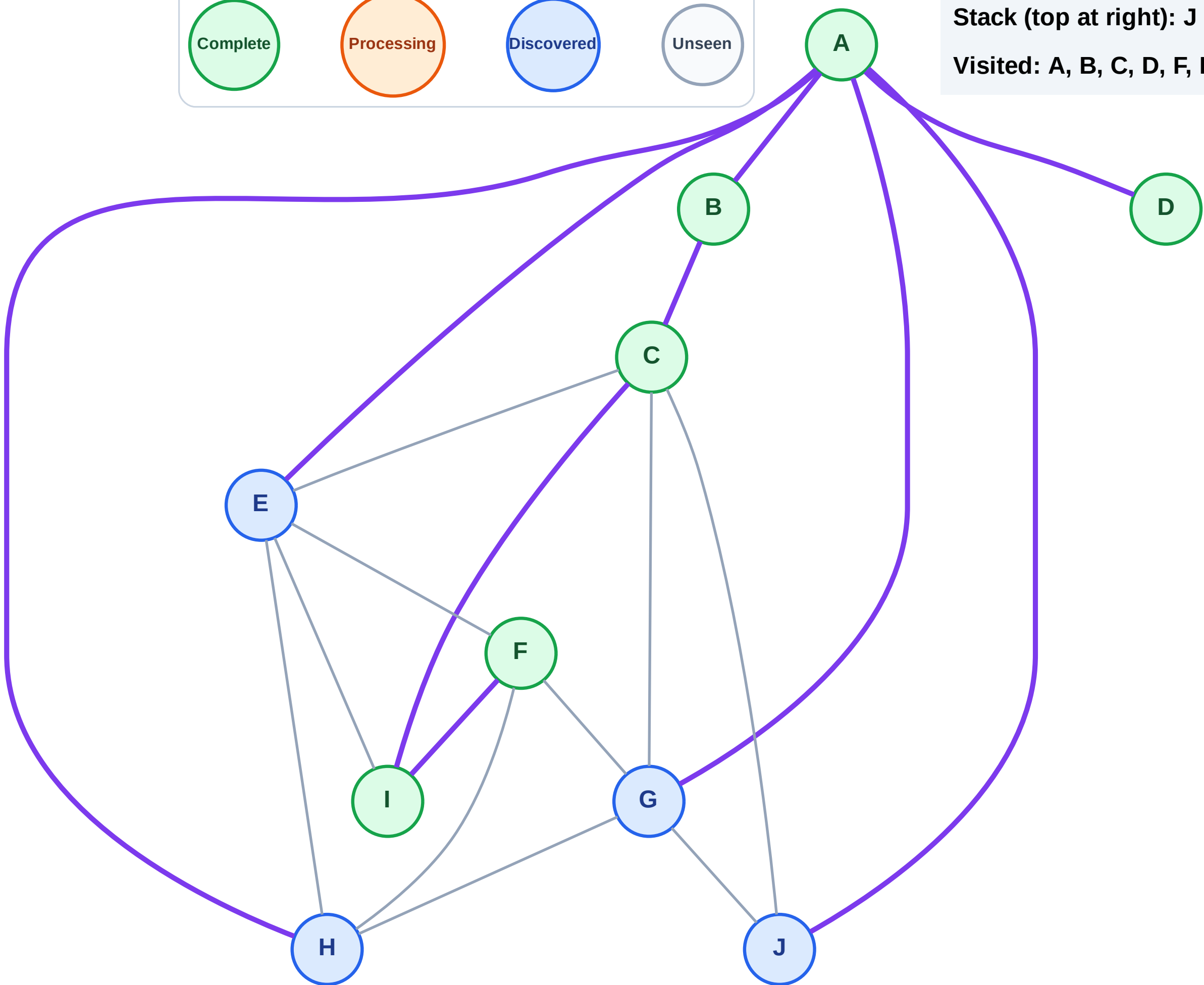


Discovered



Unseen

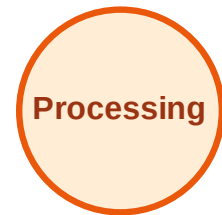
Stack (top at right): J | H | G | E
Visited: A, B, C, D, F, I



DFS Traversal

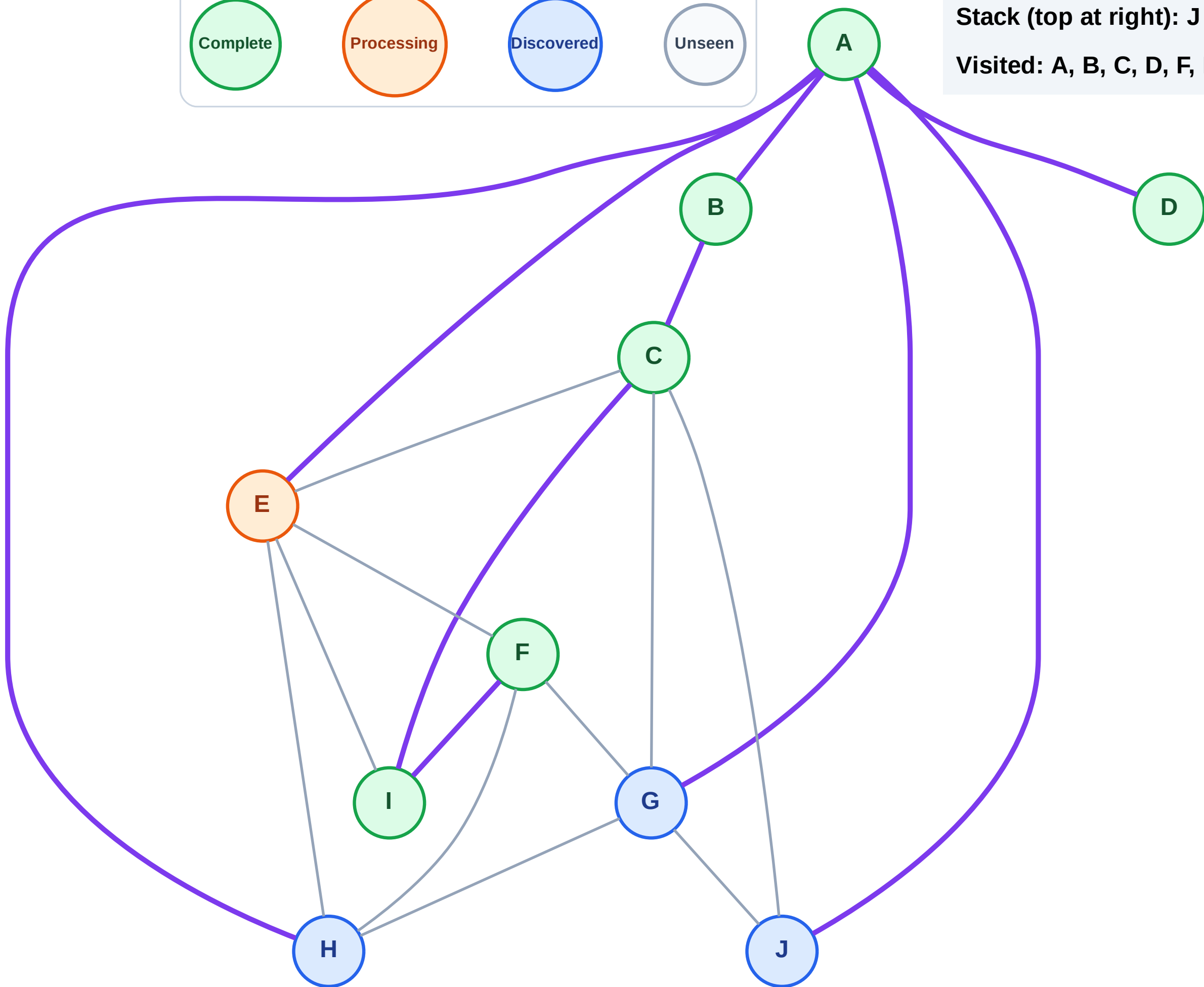
Step 14: Process node E

Legend



Stack (top at right): J | H | G

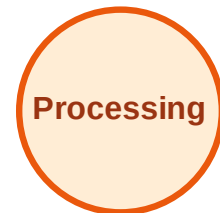
Visited: A, B, C, D, F, I



DFS Traversal

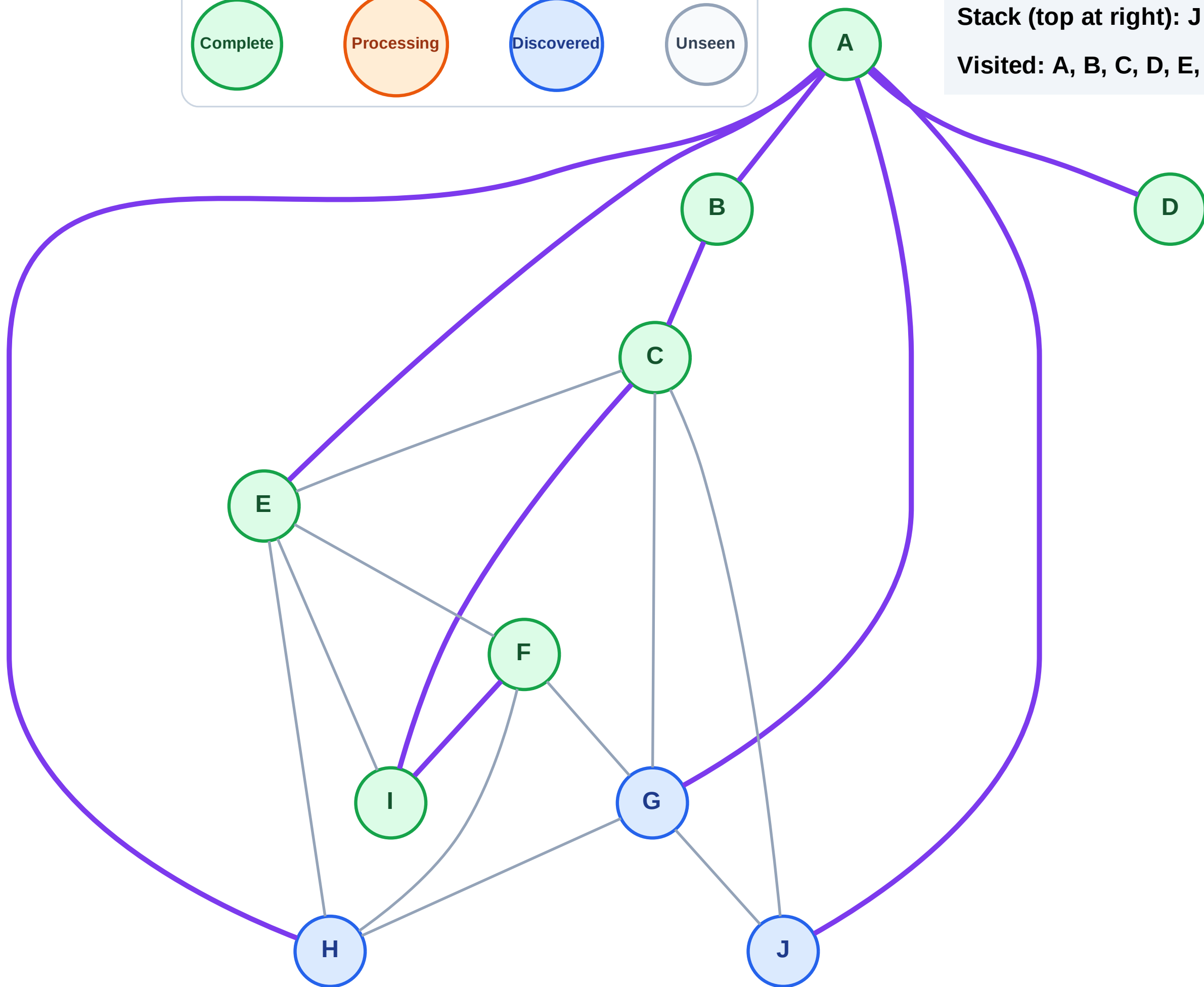
Step 15: No new neighbors from E

Legend



Stack (top at right): J | H | G

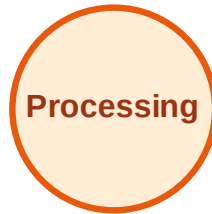
Visited: A, B, C, D, E, F, I



DFS Traversal

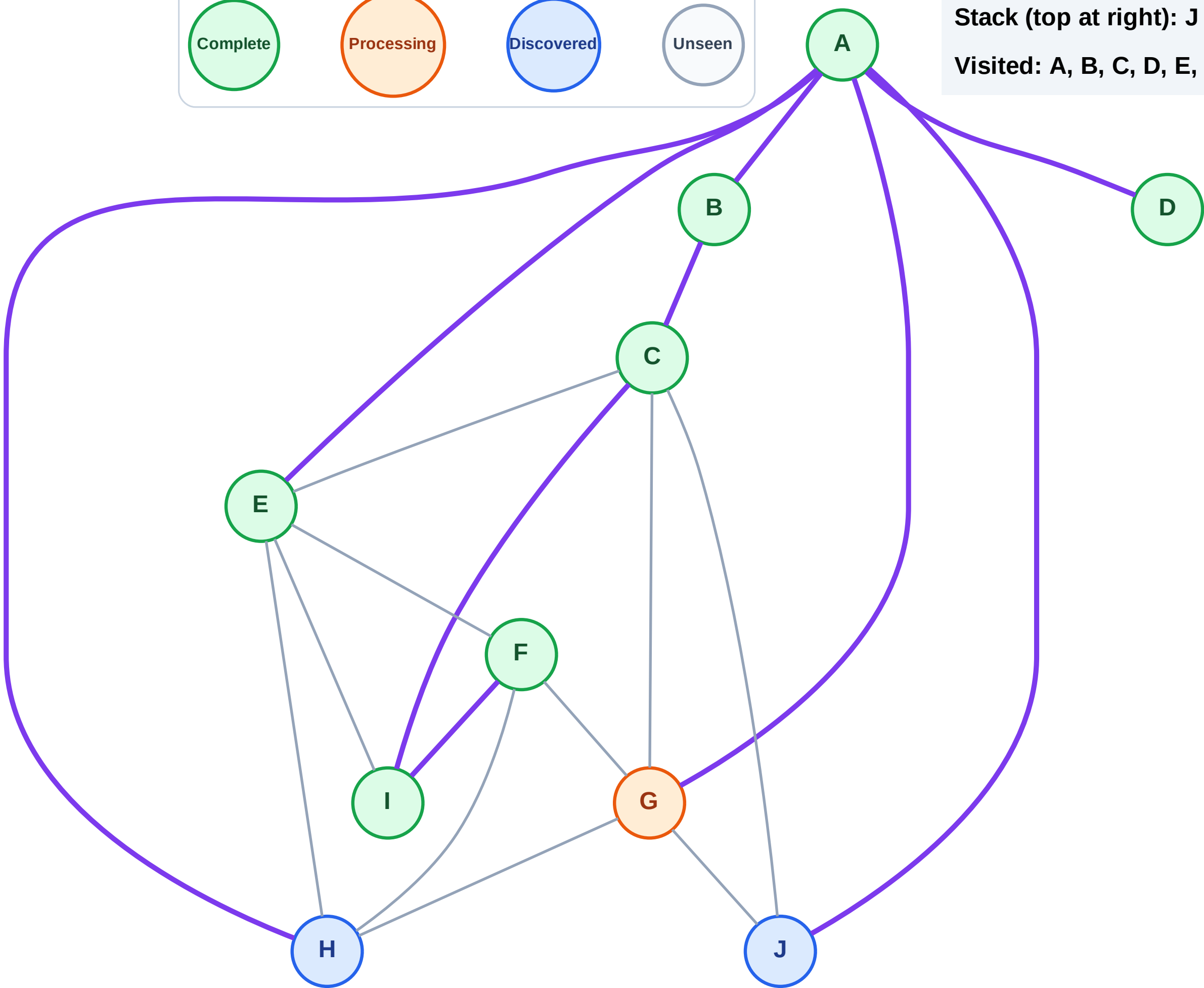
Step 16: Process node G

Legend



Stack (top at right): J | H

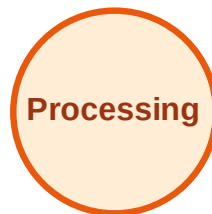
Visited: A, B, C, D, E, F, I



DFS Traversal

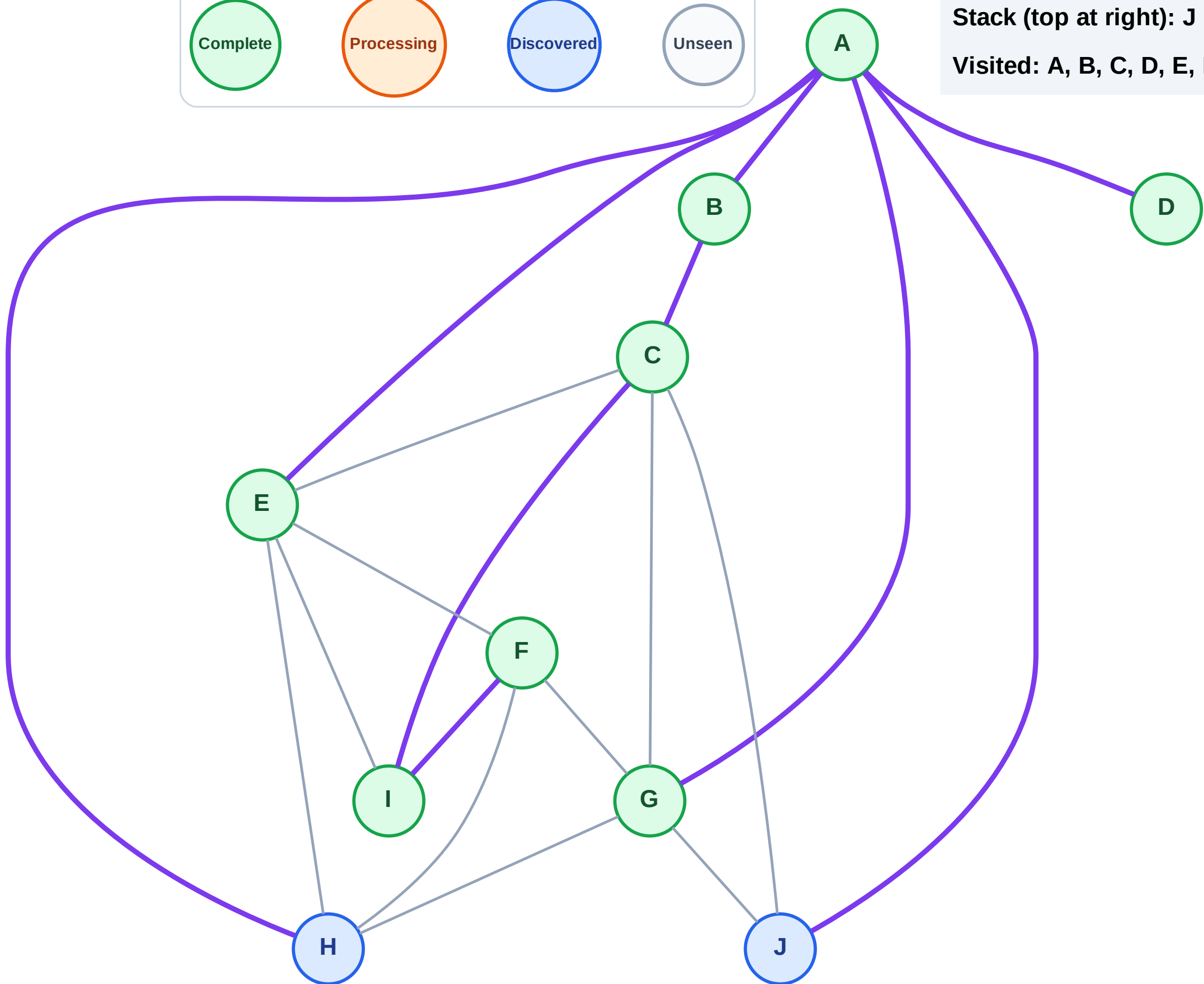
Step 17: No new neighbors from G

Legend



Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



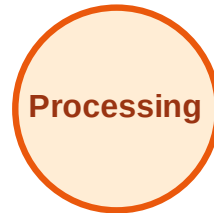
DFS Traversal

Step 18: Process node H

Legend



Complete



Processing



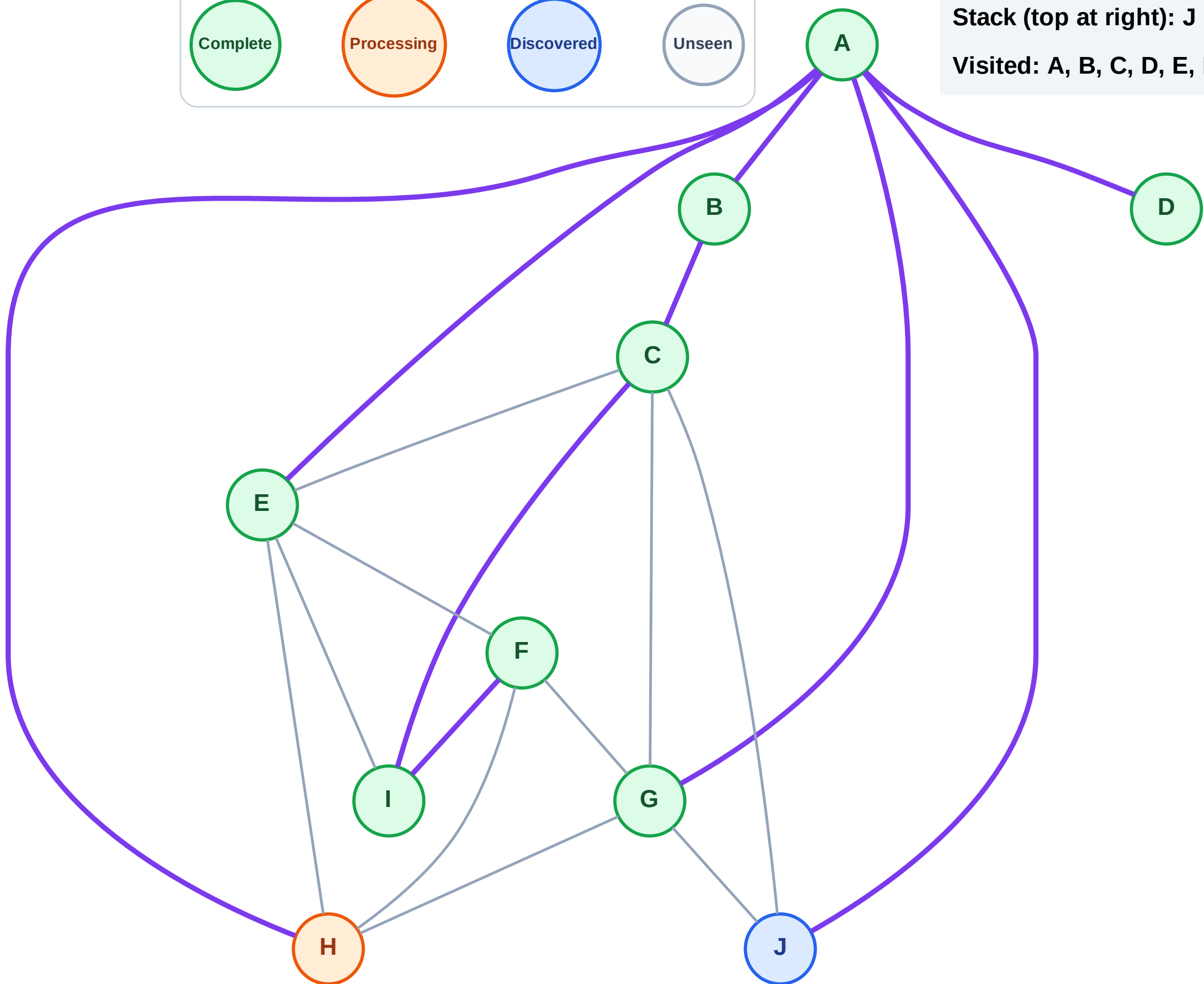
Discovered



Unseen

Stack (top at right): J

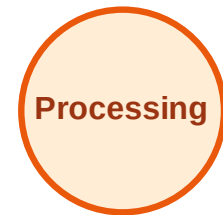
Visited: A, B, C, D, E, F, G, I



DFS Traversal

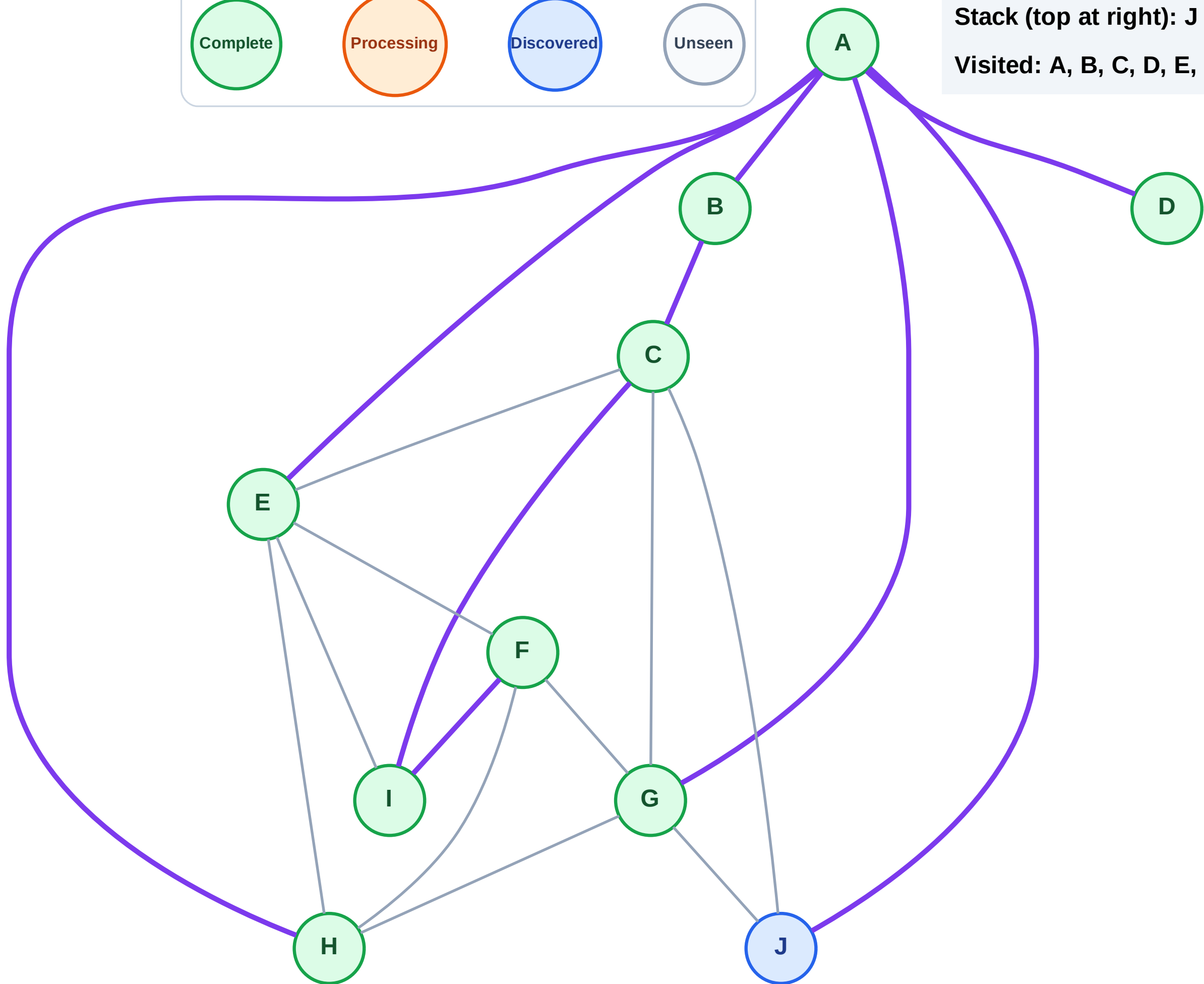
Step 19: No new neighbors from H

Legend



Stack (top at right): J

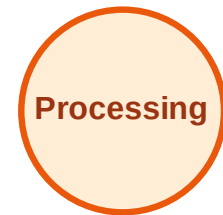
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

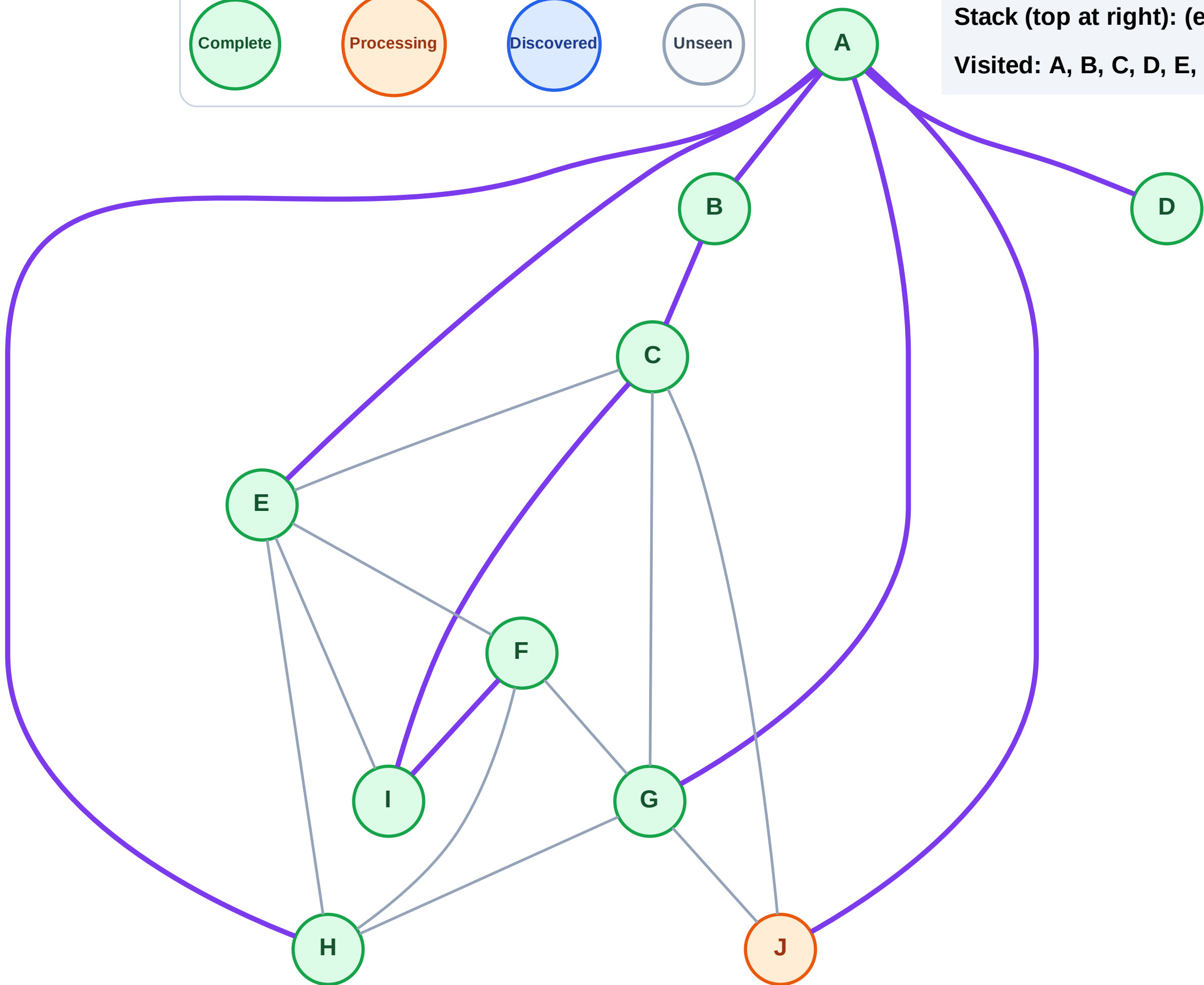
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



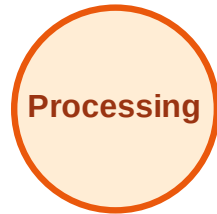
DFS Traversal

Step 21: No new neighbors from J

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

